

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-Q

(Mark One)



QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the quarterly period ended June 30, 2023

Or



TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from

to

Commission File Number 001-36198

INTERCONTINENTAL EXCHANGE, INC.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of
incorporation or organization)

46-2286804

(IRS Employer
Identification Number)

**5660 New Northside Drive,
Atlanta, Georgia**

(Address of principal executive offices)

30328

(Zip Code)

(770) 857-4700

Registrant's telephone number, including area code

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, \$0.01 par value per share	ICE	New York Stock Exchange

Indicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports); and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (\$232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company or an emerging growth company. See definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of August 1, 2023, the number of shares of the registrant's Common Stock outstanding was 560,301,373 shares.

INTERCONTINENTAL EXCHANGE, INC.

Form 10-Q

Quarterly Period Ended June 30, 2023

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PART I. Financial Statements

Item 1. Consolidated Financial Statements

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Balance Sheets
(In millions, except per share amounts)

	As of June 30, 2023 (Unaudited)	As of December 31, 2022
Assets:		
Current assets:		
Cash and cash equivalents	\$ 2,877	\$ 1,799
Short-term restricted cash and cash equivalents	5,413	6,149
Restricted short-term investments	735	—
Cash and cash equivalent margin deposits and guaranty funds	86,917	141,990
Invested deposits, delivery contracts receivable and unsettled variation margin	1,412	5,382
Customer accounts receivable, net of allowance for doubtful accounts of \$25 and \$22 at June 30, 2023 and December 31, 2022, respectively	1,313	1,169
Prepaid expenses and other current assets	555	458
Total current assets	99,222	156,947
Property and equipment, net	1,718	1,767
Other non-current assets:		
Goodwill	21,134	21,111
Other intangible assets, net	12,814	13,090
Long-term restricted cash and cash equivalents	205	405
Long-term restricted investments	199	—
Other non-current assets	991	1,018
Total other non-current assets	35,343	35,624
Total assets	\$ 136,283	\$ 194,338
Liabilities and Equity:		
Current liabilities:		
Accounts payable and accrued liabilities	\$ 919	\$ 866
Section 31 fees payable	173	223
Accrued salaries and benefits	226	352
Deferred revenue	437	170
Short-term debt	—	4
Margin deposits and guaranty funds	86,917	141,990
Invested deposits, delivery contracts payable and unsettled variation margin	1,412	5,382
Other current liabilities	120	184
Total current liabilities	90,204	149,171
Non-current liabilities:		
Non-current deferred tax liability, net	3,256	3,493
Long-term debt	18,128	18,118
Accrued employee benefits	156	160
Non-current operating lease liability	218	254
Other non-current liabilities	432	381
Total non-current liabilities	22,190	22,406
Total liabilities	112,394	171,577
Commitments and contingencies		

Equity:

Intercontinental Exchange, Inc. stockholders' equity:		
Preferred stock, \$0.01 par value; 100 shares authorized; none issued or outstanding	—	—
Common stock, \$0.01 par value; 1,500 shares authorized; 636 and 634 issued at June 30, 2023 and December 31, 2022, respectively, and 560 and 559 shares outstanding at June 30, 2023 and December 31, 2022, respectively	6	6
Treasury stock, at cost; 76 and 75 shares at June 30, 2023 and December 31, 2022, respectively	(6,276)	(6,225)
Additional paid-in capital	14,449	14,313
Retained earnings	15,925	14,943
Accumulated other comprehensive loss	(284)	(331)
Total Intercontinental Exchange, Inc. stockholders' equity	23,820	22,706
Non-controlling interest in consolidated subsidiaries	69	55
Total equity	23,889	22,761
Total liabilities and equity	\$ 136,283	\$ 194,338

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Income
(In millions, except per share amounts)
(Unaudited)

	Six Months Ended June 30,		Three Months Ended June 30,	
	2023	2022	2023	2022
Revenues:				
Exchanges	\$ 3,214	\$ 3,247	\$ 1,541	\$ 1,604
Fixed income and data services	1,109	1,021	546	512
Mortgage technology	485	604	249	297
Total revenues	4,808	4,872	2,336	2,413
Transaction-based expenses:				
Section 31 fees	175	174	56	123
Cash liquidity payments, routing and clearing	849	985	392	476
Total revenues, less transaction-based expenses	3,784	3,713	1,888	1,814
Operating expenses:				
Compensation and benefits	703	714	351	355
Professional services	57	69	29	35
Acquisition-related transaction and integration costs	46	62	25	53
Technology and communication	345	344	173	169
Rent and occupancy	45	41	25	20
Selling, general and administrative	137	112	63	57
Depreciation and amortization	527	510	267	256
Total operating expenses	1,860	1,852	933	945
Operating income	1,924	1,861	955	869
Other income/(expense):				
Interest income	193	9	102	8
Interest expense	(351)	(264)	(175)	(161)
Other income/(expense), net	(70)	(35)	(35)	23
Total other income/(expense), net	(228)	(290)	(108)	(130)
Income before income tax expense	1,696	1,571	847	739
Income tax expense	207	338	32	173
Net income	<u>\$ 1,489</u>	<u>\$ 1,233</u>	<u>\$ 815</u>	<u>\$ 566</u>
Net income attributable to non-controlling interest	(35)	(21)	(16)	(11)
Net income attributable to Intercontinental Exchange, Inc.	<u>\$ 1,454</u>	<u>\$ 1,212</u>	<u>\$ 799</u>	<u>\$ 555</u>
Earnings per share attributable to Intercontinental Exchange, Inc. common stockholders:				
Basic	<u>\$ 2.60</u>	<u>\$ 2.17</u>	<u>\$ 1.43</u>	<u>\$ 0.99</u>
Diluted	<u>\$ 2.59</u>	<u>\$ 2.16</u>	<u>\$ 1.42</u>	<u>\$ 0.99</u>
Weighted average common shares outstanding:				
Basic	<u>560</u>	<u>560</u>	<u>560</u>	<u>558</u>
Diluted	<u>561</u>	<u>562</u>	<u>561</u>	<u>560</u>

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Comprehensive Income
(In millions)
(Unaudited)

	Six Months Ended June 30,		Three Months Ended June 30,	
	2023	2022	2023	2022
Net income	\$ 1,489	\$ 1,233	\$ 815	\$ 566
Other comprehensive income/(loss):				
Foreign currency translation adjustments	47	(109)	31	(84)
Other comprehensive income/(loss)	47	(109)	31	(84)
Comprehensive income	\$ 1,536	\$ 1,124	\$ 846	\$ 482
Comprehensive income attributable to non-controlling interest	(35)	(21)	(16)	(11)
Comprehensive income attributable to Intercontinental Exchange, Inc.	\$ 1,501	\$ 1,103	\$ 830	\$ 471

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)		
	Shares	Value	Shares	Value					
Balance, as of December 31, 2022	634	\$ 6	(75)	\$ (6,225)	\$ 14,313	\$ 14,943	\$ (331)	\$ 55	\$ 22,761
Other comprehensive income	—	—	—	—	—	—	47	—	47
Exercise of common stock options	—	—	—	—	24	—	—	—	24
Payments relating to treasury shares	—	—	(1)	(51)	—	—	—	—	(51)
Stock-based compensation	—	—	—	—	94	—	—	—	94
Issuance under the employee stock purchase plan	—	—	—	—	18	—	—	—	18
Contribution from equity partners	—	—	—	—	—	—	—	9	9
Issuance of restricted stock	2	—	—	—	—	—	—	—	—
Distributions of profits	—	—	—	—	—	—	—	(30)	(30)
Dividends paid to stockholders	—	—	—	—	—	(472)	—	—	(472)
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(35)	—	35	—
Net income	—	—	—	—	—	1,489	—	—	1,489
Balance, as of June 30, 2023	636	\$ 6	(76)	\$ (6,276)	\$ 14,449	\$ 15,925	\$ (284)	\$ 69	\$ 23,889

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)		
	Shares	Value	Shares	Value					
Balance, as of March 31, 2023	635	\$ 6	(76)	\$ (6,274)	\$ 14,388	\$ 15,362	\$ (315)	\$ 44	\$ 23,211
Other comprehensive income	—	—	—	—	—	—	31	—	31
Exercise of common stock options	—	—	—	—	14	—	—	—	14
Payments relating to treasury shares	—	—	—	(2)	—	—	—	—	(2)
Stock-based compensation	—	—	—	—	47	—	—	—	47
Contribution from equity partners	—	—	—	—	—	—	—	9	9
Issuance of restricted stock	1	—	—	—	—	—	—	—	—
Dividends paid to stockholders	—	—	—	—	—	(236)	—	—	(236)
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(16)	—	16	—
Net income	—	—	—	—	—	815	—	—	815
Balance, as of June 30, 2023	636	\$ 6	(76)	\$ (6,276)	\$ 14,449	\$ 15,925	\$ (284)	\$ 69	\$ 23,889

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest - (Continued)
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)		
	Shares	Value	Shares	Value					
Balance, as of December 31, 2021	631	\$ 6	(70)	\$ (5,520)	\$ 14,069	\$ 14,350	\$ (196)	\$ 39	\$ 22,748
Other comprehensive loss	—	—	—	—	—	—	(109)	—	(109)
Exercise of common stock options	—	—	—	—	20	—	—	—	20
Repurchases of common stock	—	—	(4)	(632)	—	—	—	—	(632)
Payments relating to treasury shares	—	—	(1)	(71)	—	—	—	—	(71)
Stock-based compensation	—	—	—	—	88	—	—	—	88
Issuance under the employee stock purchase plan	—	—	—	—	24	—	—	—	24
Issuance of restricted stock	2	—	—	—	—	—	—	—	—
Distributions of profits	—	—	—	—	—	—	—	(13)	(13)
Dividends paid to stockholders	—	—	—	—	—	(427)	—	—	(427)
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(21)	—	21	—
Net income	—	—	—	—	—	1,233	—	—	1,233
Balance, as of June 30, 2022	633	\$ 6	(75)	\$ (6,223)	\$ 14,201	\$ 15,135	\$ (305)	\$ 47	\$ 22,861

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income/(Loss)		
	Shares	Value	Shares	Value					
Balance, as of March 31, 2022	633	\$ 6	(74)	\$ (6,064)	\$ 14,153	\$ 14,793	\$ (221)	\$ 36	\$ 22,703
Other comprehensive loss	—	—	—	—	—	—	(84)	—	(84)
Exercise of common stock options	—	—	—	—	5	—	—	—	5
Repurchases of common stock	—	—	(1)	(157)	—	—	—	—	(157)
Payments relating to treasury shares	—	—	—	(2)	—	—	—	—	(2)
Stock-based compensation	—	—	—	—	43	—	—	—	43
Dividends paid to stockholders	—	—	—	—	—	(213)	—	—	(213)
Net income/(loss) attributable to non-controlling interest	—	—	—	—	—	(11)	—	11	—
Net income	—	—	—	—	—	566	—	—	566
Balance, as of June 30, 2022	633	\$ 6	(75)	\$ (6,223)	\$ 14,201	\$ 15,135	\$ (305)	\$ 47	\$ 22,861

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Cash Flows
(In millions)
(Unaudited)

	Six Months Ended June 30,	
	2023	2022
Operating activities:		
Net income	\$ 1,489	\$ 1,233
Adjustments to reconcile net income to net cash provided by operating activities:		
Depreciation and amortization	527	510
Stock-based compensation	85	76
Deferred taxes	(239)	(147)
Gain on sale of Euroclear investment	—	(41)
Net losses from unconsolidated investees	65	57
Other	26	21
Changes in assets and liabilities:		
Customer accounts receivable	(181)	(177)
Other current and non-current assets	(66)	(48)
Section 31 fees payable	(50)	115
Deferred revenue	279	292
Other current and non-current liabilities	(130)	(166)
Total adjustments	316	492
Net cash provided by operating activities	1,805	1,725
Investing activities:		
Capital expenditures	(61)	(70)
Capitalized software development costs	(142)	(134)
Purchases of invested margin deposits	(771)	(1,431)
Proceeds from sales of invested margin deposits	3,078	3,815
Cash paid for acquisitions, net of cash acquired	(5)	(36)
Purchases of equity and equity method investments	(6)	(43)
Proceeds from sale of Euroclear investment	—	741
Purchases of investments	(949)	—
Other	—	1
Net cash provided by investing activities	1,144	2,843
Financing activities:		
Proceeds from/(repayments of) debt facilities, net	(4)	5,186
Redemption of commercial paper, net	—	(1,012)
Capital contributions from non-controlling interests	9	—
Repurchases of common stock	—	(632)
Dividends to stockholders	(472)	(427)
Change in cash and cash equivalent margin deposits and guaranty funds	(57,380)	16,163
Payments relating to treasury shares received for restricted stock tax payments and stock option exercises	(50)	(71)
Other	11	31
Net cash provided by/(used in) financing activities	(57,886)	19,238
Effect of exchange rate changes on cash, cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds	6	(19)
Net increase/(decrease) in cash, cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds	(54,931)	23,787
Cash, cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds at beginning of period	150,343	147,976
Cash, cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds at end of period	\$ 95,412	\$ 171,763

	As of June 30, 2023	As of June 30, 2022
Supplemental cash flow disclosure:		
Cash paid for income taxes	\$ 536	\$ 524
Cash paid for interest	\$ 341	\$ 226
Reconciliation of the components of cash, cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds to the balance sheet:		
Cash and cash equivalents	\$ 2,877	\$ 830
Short-term restricted cash and cash equivalents	5,413	6,045
Long-term restricted cash and cash equivalents	205	405
Cash and cash equivalent margin deposits and guaranty funds	86,917	164,483
Total	\$ 95,412	\$ 171,763

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Notes to Consolidated Financial Statements
(Unaudited)

1. Description of Business

Nature of Business and Organization

Intercontinental Exchange, Inc. is a provider of market infrastructure, data services and technology solutions to a broad range of customers including financial institutions, corporations and government entities. These products, which span major asset classes including futures, equities, fixed income and United States, or U.S., residential mortgages provide our customers with access to mission critical tools that are designed to increase asset class transparency and workflow efficiency. Our business is conducted through three reportable business segments:

- **Exchanges:** We operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities.
- **Fixed Income and Data Services:** We provide fixed income pricing, reference data, indices, analytics and execution services as well as global credit default swap, or CDS, clearing and multi-asset class data delivery solutions.
- **Mortgage Technology:** We provide a technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

We operate marketplaces, technology and provide data services in the U.S., United Kingdom, or U.K., European Union, or EU, Canada, Asia Pacific and the Middle East.

2. Summary of Significant Accounting Policies

Basis of Presentation

The accompanying unaudited consolidated financial statements have been prepared in accordance with U.S. generally accepted accounting principles, or U.S. GAAP, pursuant to the rules and regulations of the Securities and Exchange Commission, or SEC, regarding interim financial reporting. Accordingly, the unaudited consolidated financial statements do not include all of the information and footnotes required by U.S. GAAP for complete financial statements and should be read in conjunction with our audited consolidated financial statements and related notes thereto for the year ended December 31, 2022. The accompanying unaudited consolidated financial statements reflect all adjustments that are, in our opinion, necessary for a fair presentation of results for the interim periods presented. We believe that these adjustments are of a normal recurring nature.

Preparing financial statements in conformity with U.S. GAAP requires us to make certain estimates and assumptions that affect the amounts that are reported in our consolidated financial statements and accompanying disclosures. Actual amounts could differ from those estimates. The results of operations for the six months and three months ended June 30, 2023 are not necessarily indicative of the results to be expected for any future period or the full fiscal year.

These statements include the accounts of our wholly-owned and controlled subsidiaries. All intercompany balances and transactions between us and our wholly-owned and controlled subsidiaries have been eliminated in consolidation. For consolidated subsidiaries in which our ownership is less than 100% and for which we have control over the assets and liabilities and the management of the entity, the outside stockholders' interests are shown as non-controlling interests.

We have considered the impacts of macroeconomic conditions during the quarter, including interest rates, the inflationary environment, geopolitical events and military conflicts, including repercussions from the conflict in Ukraine, and the impact that any of the foregoing may have on the global economy and on our business. As of June 30, 2023, our businesses and operations, including our exchanges, clearing houses, listings venues, data services businesses and mortgage platforms, have not suffered a material negative impact as a result of these events. There continues to be uncertainty surrounding the current macroeconomic environment and the impact that it may have on the global economy and on our business.

Recently Adopted Accounting Pronouncements

During the six months ended June 30, 2023, there were no significant changes to the new and recently adopted accounting pronouncements applicable to us from those disclosed in Note 2 to the consolidated financial statements in Part II, Item 8 of our Annual Report on Form 10-K for the year ended December 31, 2022, or the 2022 Form 10-K.

3. Pending Acquisition

Pending Acquisition of Black Knight, Inc.

On May 4, 2022, we announced that we had entered into a definitive agreement to acquire Black Knight, Inc., or Black Knight, a software, data and analytics company that serves the housing finance continuum, including real estate data, mortgage lending and servicing, as well as the secondary markets. Pursuant to that certain Agreement and Plan of Merger, dated as of May 4, 2022, among ICE, Sand Merger Sub Corporation, a wholly owned subsidiary of ICE, or Sub, and Black Knight, which we refer to as the “merger agreement,” Sub will merge with and into Black Knight, which we refer to as the “merger,” with Black Knight surviving as a wholly owned subsidiary of ICE.

On March 7, 2023, ICE and Black Knight announced that, in connection with the merger agreement, Black Knight has entered into an agreement to sell its Empower loan origination business. On March 7, 2023, ICE and Black Knight also entered into an amendment to the merger agreement to reduce the value of the aggregate transaction consideration to approximately \$11.7 billion as of March 7, 2023, or \$75 per share of Black Knight common stock, with cash comprising 90% of the value of the aggregate transaction consideration and shares of our common stock comprising 10% of the value of the aggregate transaction consideration. The aggregate cash component of the transaction consideration is fixed at \$10.5 billion, and the value of the aggregate stock component of the transaction consideration will fluctuate with the market price of our common stock and will be determined based on the average of the volume weighted averages of the trading prices of our common stock on each of the ten consecutive trading days ending three trading days prior to the closing of the merger. If consummated, we expect that this transaction will build on our position as a provider of end-to-end electronic workflow solutions for the rapidly evolving U.S. residential mortgage industry. We believe the Black Knight ecosystem adds value for clients of all sizes across the mortgage and real estate lifecycles by helping organizations lower costs, increase efficiencies, grow their businesses, and reduce risk.

On March 30, 2023, our amended proxy statement/prospectus on Form S-4 was declared effective by the SEC, and on April 28, 2023, Black Knight stockholders approved the amendment to the merger agreement.

On July 17, 2023, we announced that we have agreed to sell Black Knight’s Optimal Blue business to the same purchaser as that of the Empower business for \$700 million. The structure of this proposed transaction includes a payment of \$200 million in cash, with the remainder financed by a \$500 million promissory note issued by the purchaser to Black Knight, as a subsidiary of ICE, at the closing of the transaction. The closing of the proposed Optimal Blue divestiture transaction is subject to the closing of our acquisition of Black Knight, the closing of the acquisition of Black Knight’s Empower loan origination system business, and the satisfaction of other customary closing conditions.

The transaction is expected to close in the second half of 2023 following the receipt of regulatory approvals, a favorable resolution of the FTC litigation concerning this transaction, and the satisfaction of customary closing conditions. See Note 13 where additional details of this transaction are discussed.

4. Investments

Equity Investments Subject to ASU 2016-01

Our equity investments are subject to valuation under Accounting Standards Update, or ASU, 2016-01, *Financial Instruments- Overall (Subtopic 825-10): Recognition and Measurement of Financial Assets and Financial Liabilities*, or ASU 2016-01. See Note 14 for a discussion of our determination of fair value of our financial instruments, which were not material as of June 30, 2023.

Investment in Euroclear

We previously owned a 9.8% stake in Euroclear, plc, or Euroclear, that we originally purchased for \$631 million. We participated on the Euroclear Board of Directors, and we classified our investment in Euroclear as an equity investment.

During the three months ended June 30, 2022, we completed the sale of our 9.8% stake in Euroclear. The carrying value of our investment was \$700 million at the time of the sale and was classified within other current assets on our balance sheet. We recorded a net gain on the sale of \$41 million, which was included in other income, during the three months ended June 30, 2022.

Equity Method Investments

Our equity method investments include the Options Clearing Corporation, or OCC, and Bakkt Holdings, LLC, or Bakkt, among others. Our equity method investments are included in other non-current assets in the accompanying consolidated

balance sheet. We initially record our equity method investments at cost. At the end of each reporting period, we record our share of profits or losses of our equity method investments as equity earnings included in other income, and adjust the carrying value of our equity method investment accordingly. In addition, if and when our equity method investments issue cash dividends to us, we deduct the amount of these dividends from the carrying amount of that investment. We assess the carrying value periodically if impairment indicators are present.

We recognized \$65 million and \$57 million as our share of estimated losses, net, from our equity method investments during the six months ended June 30, 2023 and 2022, respectively, and \$30 million and \$15 million as our share of losses, net, from our equity method investments during the three months ended June 30, 2023 and 2022, respectively. The estimated losses during both the six months and three months ended June 30, 2023 and 2022 are primarily related to our investment in Bakkt, partially offset by our share of OCC profits. Both periods include adjustments to reflect the difference between reported prior period actual results from our original estimates.

When performing our assessment of the carrying value of our investments, we consider, among other things, the length of time and the extent to which the market value has been less than our cost basis, if applicable, the investee's financial condition and near-term prospects, the economic or technological environment in which our investees operate, weakening of the general market condition of the related industry, whether an investee can continue as a going concern, any impairment charges recorded by an investee on goodwill, intangible or long-lived assets, and our intent and ability to retain the investment for a period of time sufficient to allow for any anticipated recovery in market value.

Investment in OCC

We own a 40% interest in OCC through a direct investment by the New York Stock Exchange, or NYSE. OCC is regulated by the SEC as a registered clearing agency and by the Commodity Futures Trading Commission, or CFTC, as a derivatives clearing organization. OCC serves as a clearing house for securities options, security futures, commodity futures and options on futures traded on various independent exchanges. OCC clears securities options traded on NYSE Arca and NYSE Amex Options, along with other non-affiliated exchanges.

Investment in Bakkt

As of June 30, 2023, we held an approximate 64% economic interest in Bakkt. As a result of limitations on ICE from the Bakkt voting agreement entered into in connection with Bakkt's merger with VIH, we hold a minority voting interest in Bakkt and treat it as an equity method investment.

5. Revenue Recognition

Substantially all of our revenues are considered to be revenues from contracts with customers. The related accounts receivable balances are recorded in our balance sheets as customer accounts receivable. We do not have obligations for warranties, returns or refunds to customers, other than rebates, which are settled each period and therefore do not result in variable consideration. We do not have significant revenue recognized from performance obligations that were satisfied in prior periods, and we do not have any transaction price allocated to unsatisfied performance obligations other than in our deferred revenue. Certain judgments and estimates are used in the identification and timing of satisfaction of performance obligations and the related allocation of transaction price. We believe that these represent a faithful depiction of the transfer of services to our customers.

Deferred revenue represents our contract liabilities related to our annual, original and other listings revenues, certain data services, clearing services, mortgage technology services and other revenues. Deferred revenue is our only significant contract liability. See Note 7 for our discussion of deferred revenue balances, activity, and expected timing of recognition.

For all of our contracts with customers, except for listings and certain data, clearing and mortgage services, our performance obligations are short term in nature and there is no significant variable consideration. In addition, we have elected the practical expedient of excluding sales taxes from transaction prices. We have assessed the costs incurred to obtain or fulfill a contract with a customer, which are primarily our sales commissions.

Refer to Note 5 to the consolidated financial statements included in Part II, Item 8 of our 2022 Form 10-K where our primary revenue contract classifications are described in detail.

The following table depicts the disaggregation of our revenue according to business line and segment (in millions). Amounts here have been aggregated as they follow consistent revenue recognition patterns, and are consistent with the segment information in Note 15:

	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Six Months Ended June 30, 2023:				
Total revenues	\$ 3,214	\$ 1,109	\$ 485	\$ 4,808
Transaction-based expenses	1,024	—	—	1,024
Total revenues, less transaction-based expenses	<u>\$ 2,190</u>	<u>\$ 1,109</u>	<u>\$ 485</u>	<u>\$ 3,784</u>

Timing of Revenue Recognition

Services transferred at a point in time	\$ 1,263	\$ 229	\$ 148	\$ 1,640
Services transferred over time	927	880	337	2,144
Total revenues, less transaction-based expenses	<u>\$ 2,190</u>	<u>\$ 1,109</u>	<u>\$ 485</u>	<u>\$ 3,784</u>

	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Three Months Ended June 30, 2023:				
Total revenues	\$ 1,541	\$ 546	\$ 249	\$ 2,336
Transaction-based expenses	448	—	—	448
Total revenues, less transaction-based expenses	<u>\$ 1,093</u>	<u>\$ 546</u>	<u>\$ 249</u>	<u>\$ 1,888</u>

Timing of Revenue Recognition

Services transferred at a point in time	\$ 628	\$ 106	\$ 81	\$ 815
Services transferred over time	465	440	168	1,073
Total revenues, less transaction-based expenses	<u>\$ 1,093</u>	<u>\$ 546</u>	<u>\$ 249</u>	<u>\$ 1,888</u>

	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Six Months Ended June 30, 2022:				
Total revenues	\$ 3,247	\$ 1,021	\$ 604	\$ 4,872
Transaction-based expenses	1,159	—	—	1,159
Total revenues, less transaction-based expenses	<u>\$ 2,088</u>	<u>\$ 1,021</u>	<u>\$ 604</u>	<u>\$ 3,713</u>

Timing of Revenue Recognition

Services transferred at a point in time	\$ 1,197	\$ 159	\$ 270	\$ 1,626
Services transferred over time	891	862	334	2,087
Total revenues, less transaction-based expenses	<u>\$ 2,088</u>	<u>\$ 1,021</u>	<u>\$ 604</u>	<u>\$ 3,713</u>

	Exchanges Segment	Fixed Income and Data Services Segment	Mortgage Technology Segment	Total Consolidated
Three Months Ended June 30, 2022:				
Total revenues	\$ 1,604	\$ 512	\$ 297	\$ 2,413
Transaction-based expenses	599	—	—	599
Total revenues, less transaction-based expenses	<u>\$ 1,005</u>	<u>\$ 512</u>	<u>\$ 297</u>	<u>\$ 1,814</u>
Timing of Revenue Recognition				
Services transferred at a point in time	\$ 563	\$ 83	\$ 128	\$ 774
Services transferred over time	442	429	169	1,040
Total revenues, less transaction-based expenses	<u>\$ 1,005</u>	<u>\$ 512</u>	<u>\$ 297</u>	<u>\$ 1,814</u>

The Exchanges segment and the Fixed Income and Data Services segment revenues above include data services revenues. Our data services revenues are transferred over time, and a majority of those revenues are performed over a short period of time of one month or less and relate to subscription-based data services billed monthly, quarterly or annually in advance. These revenues are recognized ratably over time as our data delivery performance obligations are met consistently throughout the period.

The Exchanges segment revenues transferred over time in the table above also include services related to listings, services related to risk management of open interest performance obligations and services related to regulatory fees, trading permits, and software licenses.

The Fixed Income and Data Services segment revenues transferred over time in the table above also include services related to risk management of open interest performance obligations, primarily in our CDS business.

The Mortgage Technology segment revenues transferred over time in the table above primarily relate to our origination technology revenue where performance obligations consist of a series of distinct services and are recognized over the contract terms as subscription performance obligations are satisfied, and to a lesser extent, professional services revenues and revenues from certain of our data and analytics offerings.

The components of services transferred over time for each of our segments are as follows:

	Six Months Ended June 30,		Three Months Ended June 30,	
	2023	2022	2023	2022
Exchanges Segment:				
Data services revenues	\$ 463	\$ 432	\$ 231	\$ 218
Services transferred over time related to risk management of open interest performance obligations	151	139	\$ 75	\$ 63
Services transferred over time related to listings	252	260	\$ 126	\$ 131
Services transferred over time related to regulatory fees, trading permits, and software licenses	61	60	\$ 33	\$ 30
Total	<u>\$ 927</u>	<u>\$ 891</u>	<u>\$ 465</u>	<u>\$ 442</u>
Fixed Income Data Services Segment:				
Data services revenues	\$ 864	\$ 843	\$ 434	\$ 421
Services transferred over time related to risk management of open interest performance obligations in our CDS business	16	19	\$ 6	\$ 8
Total	<u>\$ 880</u>	<u>\$ 862</u>	<u>\$ 440</u>	<u>\$ 429</u>
Mortgage Technology Segment:				
Subscription revenues	\$ 329	\$ 316	\$ 164	\$ 160
Professional service revenues and other	8	18	\$ 4	\$ 9
Total	<u>\$ 337</u>	<u>\$ 334</u>	<u>\$ 168</u>	<u>\$ 169</u>
Total consolidated revenues transferred over time	<u>\$ 2,144</u>	<u>\$ 2,087</u>	<u>\$ 1,073</u>	<u>\$ 1,040</u>

6. Goodwill and Other Intangible Assets

The following is a summary of the activity in our goodwill balance for the six months ended June 30, 2023 (in millions):

Goodwill balance at December 31, 2022	\$	21,111
Foreign currency translation		22
Other activity, net		1
Goodwill balance at June 30, 2023	\$	21,134

The following is a summary of the activity in our other intangible assets balance for the six months ended June 30, 2023 (in millions):

Other intangible assets balance at December 31, 2022	\$	13,090
Acquisitions		5
Foreign currency translation		20
Amortization of other intangible assets		(301)
Other intangible assets balance at June 30, 2023	\$	12,814

Foreign currency translation adjustments result from a portion of our goodwill and other intangible assets being held at our U.K., EU and Canadian subsidiaries, whose functional currencies are not the U.S. dollar. The changes in other activity, net, in the table above primarily relate to adjustments to the fair value of the net tangible and intangible assets made within one year of acquisitions, with a corresponding adjustment to goodwill.

During the six months ended June 30, 2023, we considered potential indicators of impairment to goodwill and other intangible assets for each of our reporting units, which included continued global inflation concerns and rising interest rates, including their effect on our forecasts, among other things. As such, we performed this assessment to determine whether it was more-likely-than-not that goodwill and indefinite lived intangibles within each of our reportable business segments were impaired. Additionally, we evaluated whether the carrying value of the finite lived intangible assets within our reportable business segments may not be recoverable. After evaluating events, circumstances and factors which could affect the significant inputs used in our evaluation of cash flows and related fair value, we determined it was not more-likely-than-not that an impairment existed in our goodwill and indefinite lived intangible assets or that the carrying amount of our finite lived intangible assets was not recoverable. We plan to perform our annual impairment testing in the fourth quarter.

7. Deferred Revenue

Our contract liabilities, or deferred revenue, represent consideration received that is yet to be recognized as revenue. Total deferred revenue was \$529 million as of June 30, 2023, including \$437 million in current deferred revenue and \$92 million in other non-current liabilities. The changes in our deferred revenue during the six months ended June 30, 2023 are as follows (in millions):

	Listings Revenues	Data Services and Other Revenues	Mortgage Technology	Total
Deferred revenue balance at January 1, 2023	\$ 115	\$ 88	\$ 51	\$ 254
Additions	474	236	32	742
Amortization	(252)	(175)	(40)	(467)
Deferred revenue balance at June 30, 2023	\$ 337	\$ 149	\$ 43	\$ 529

The changes in our deferred revenue during the six months ended June 30, 2022 are as follows (in millions):

	Listings Revenues	Data Services and Other Revenues	Mortgage Technology	Total
Deferred revenue balance at January 1, 2022	\$ 112	\$ 93	\$ 79	\$ 284
Additions	500	283	46	829
Amortization	(261)	(222)	(55)	(538)
Deferred revenue balance at June 30, 2022	<u>\$ 351</u>	<u>\$ 154</u>	<u>\$ 70</u>	<u>\$ 575</u>

Included in the amortization recognized during the six months ended June 30, 2023 is \$103 million related to the deferred revenue balance as of January 1, 2023. Included in the amortization recognized for the six months ended June 30, 2022 is \$119 million related to the deferred revenue balance as of January 1, 2022. As of June 30, 2023, the remaining deferred revenue balance will be recognized over the period of time we satisfy our performance obligations as described in Note 5.

8. Debt

Our total debt, including short-term and long-term debt, consisted of the following (in millions):

	As of June 30, 2023	As of December 31, 2022
Debt:		
Short-term debt:		
Other short-term debt	\$ —	\$ 4
Total short-term debt	<u>—</u>	<u>4</u>
Long-term debt:		
2025 Senior Notes (3.65% senior unsecured notes due May 23, 2025)	1,245	1,243
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)	1,248	1,247
2027 Senior Notes (4.00% senior unsecured notes due September 15, 2027)	1,488	1,487
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)	498	498
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)	595	594
2029 Senior Notes (4.35% senior unsecured notes due June 15, 2029)	1,240	1,240
2030 Senior Notes (2.10% senior unsecured notes due June 15, 2030)	1,237	1,235
2032 Senior Notes (1.85% senior unsecured notes due September 15, 2032)	1,485	1,485
2033 Senior Notes (4.60% senior unsecured notes due March 15, 2033)	1,488	1,488
2040 Senior Notes (2.65% senior unsecured notes due September 15, 2040)	1,232	1,231
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)	1,232	1,231
2050 Senior Notes (3.00% senior unsecured notes due June 15, 2050)	1,221	1,221
2052 Senior Notes (4.95% senior unsecured notes due June 15, 2052)	1,465	1,464
2060 Senior Notes (3.00% senior unsecured notes due September 15, 2060)	1,471	1,471
2062 Senior Notes (5.20% senior unsecured notes due June 15, 2062)	983	983
Total long-term debt	<u>18,128</u>	<u>18,118</u>
Total debt	<u>\$ 18,128</u>	<u>\$ 18,122</u>

Our senior notes of \$18.1 billion have a weighted average maturity of 16 years and a weighted average cost of 3.6% per annum.

Credit Facilities

We have a \$3.9 billion senior unsecured revolving credit facility, or the Credit Facility, with a maturity date of May 25, 2027, with future capacity to increase our borrowings under the Credit Facility by an additional \$1.0 billion, subject to the consent of the lenders funding the increase and certain other conditions. No amounts were outstanding under the Credit Facility as of June 30, 2023.

As of June 30, 2023, of the \$3.9 billion that was available for borrowing under the Credit Facility, \$171 million was required to support certain broker-dealer and other subsidiary commitments. We did not have any amounts outstanding under our U.S. dollar commercial paper program, or the Commercial Paper Program, as of June 30, 2023. Therefore, there was not an amount required to backstop the Commercial Paper Program. The amount required to backstop the amounts

outstanding under the Commercial Paper Program will fluctuate as we increase or decrease our commercial paper borrowings. The remaining \$3.7 billion was available for working capital and general corporate purposes including, but not limited to, acting as a backstop to future amounts outstanding under the Commercial Paper Program.

We have a \$2.4 billion two-year senior unsecured delayed draw term loan facility, or the Term Loan, with a maturity date of May 25, 2024. Draws under the Term Loan bear interest on the principal amount outstanding at either (a) Term Secured Overnight Financing Rate, or Term SOFR, plus an applicable margin plus a credit spread adjustment of 10 basis points or (b) a "base rate" plus an applicable margin. The applicable margin ranges from 0.625% to 1.125% for Term SOFR loans and from 0.000% to 0.125% for base rate loans, in each case, based on a ratings-based pricing grid. We expect to use the proceeds from borrowings under the Term Loan to fund a portion of the purchase price for the Black Knight acquisition. We have the option to prepay outstanding amounts under the Term Loan in whole or in part at any time. No amounts were outstanding under the Term Loan as of June 30, 2023.

Our India subsidiaries maintain \$14 million of credit lines for their general corporate purposes. As of June 30, 2023, there were no amounts outstanding under these credit lines.

Commercial Paper Program

Our Commercial Paper Program is currently backed by the borrowing capacity available under the Credit Facility, as described above. The effective interest rate of commercial paper issuances does not materially differ from short-term interest rates, which fluctuate due to market conditions and as a result may impact our interest expense. We did not have any notes outstanding under our Commercial Paper Program as of June 30, 2023.

9. Share-Based Compensation

We currently sponsor stock option plans, restricted stock plans and our Employee Stock Purchase Plan to our employees and directors. Stock options and restricted stock are granted at the discretion of the Compensation Committee of our Board of Directors, or Board, based on the estimated fair value on the date of grant. The fair value of the stock options and restricted stock on the date of grant is recognized as expense over the vesting period, net of forfeitures. The non-cash compensation expenses recognized in our consolidated statements of income for stock options, restricted stock and under our employee stock purchase plan, net of amounts classified as capitalized software, were \$85 million and \$76 million for the six months ended June 30, 2023 and 2022, respectively, and \$45 million and \$38 million during the three months ended June 30, 2023 and 2022, respectively.

Stock Option Plans

We use the Black-Scholes option pricing model to value our stock option awards. During the six months ended June 30, 2023 and 2022, we used the assumptions in the table below to compute the value:

Assumptions:	Six Months Ended June 30,	
	2023	2022
Risk-free interest rate	3.47%	1.72%
Expected life in years	6.1	6.0
Expected volatility	24%	23%
Expected dividend yield	1.56%	1.17%
Estimated weighted-average fair value of options granted per share	\$27.39	\$28.18

The risk-free interest rate is based on the zero-coupon U.S. Treasury yield curve in effect at the date of grant. The expected life is derived from historical and anticipated future exercise patterns. Expected volatility is based on historical volatility data of our stock.

Restricted Stock Plans

Restricted shares are used as an incentive to attract and retain qualified employees and to align our and our stockholders' interests by linking actual performance to both short and long-term stockholder return. We issue awards that may contain a combination of time, performance and/or market conditions. The grant date fair value of each award is based on the closing stock price of our stock at the date of grant. The grant date fair value of time-based restricted stock is recognized as expense ratably over the vesting period, which is typically three or four years, net of forfeitures.

In February 2023, we reserved a maximum of 0.9 million restricted shares for potential issuance as performance-based restricted shares to certain of our employees. The number of shares ultimately granted under this award will be based on our actual financial performance as compared to financial performance targets set by our Board and the Compensation Committee for the year ending December 31, 2023, and will also be subject to a market condition reduction based on how

our 2023 total stockholder return, or TSR, compares to that of the S&P 500 Index. The maximum compensation expense to be recognized under these performance-based restricted shares is \$92 million if the maximum financial performance target is met and all 0.9 million shares vest. The compensation expense to be recognized under these performance-based restricted shares will be \$46 million if the target financial performance is met, which would result in 0.4 million shares vesting. For these awards with performance conditions, we recognize expense on an accelerated basis over the three-year vesting period based on our quarterly assessment of the probable 2023 actual financial performance as compared to the 2023 financial performance targets. As of June 30, 2023, our best estimate is that the financial performance level will be above target for 2023. Based on this assessment, we recorded non-cash compensation expense of \$14 million and \$10 million for the six months and three months ended June 30, 2023, respectively, related to these awards and the remaining \$46 million in non-cash compensation expense will be recorded on an accelerated basis over the remaining vesting period, including \$19 million which will be recorded over the remainder of 2023.

We also issue awards with a market condition but no performance condition. The fair value of these awards is estimated based on a simulation of various outcomes and includes inputs such as our stock price on the grant date, the valuation of historical awards with market conditions, the relatively low likelihood that the market condition will affect the number of shares granted (as the market condition only affects shares granted in excess of certain financial performance targets), and our expectation of achieving the financial performance targets.

10. Equity

Stock Repurchase Program

In December 2021, our Board approved an aggregate of \$3.15 billion for future repurchases of our common stock with no fixed expiration date that became effective on January 1, 2022. The approval of our Board for the share repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board may increase or decrease the amount available for repurchases from time to time. We fund repurchases from our operating cash flow or borrowings under our debt facilities or our Commercial Paper Program. Repurchases may be made from time to time on the open market, through established trading plans, in privately-negotiated transactions or otherwise, in accordance with all applicable securities laws, rules and regulations. We may begin or discontinue stock repurchases at any time and may amend or terminate a Rule 10b5-1 trading plan at any time or enter into additional plans, subject to applicable rules.

We did not have any share repurchases during the six months or three months ended June 30, 2023. During the six months and three months ended June 30, 2022, we repurchased a total of 5.0 million and 1.3 million shares, respectively, of our outstanding common stock at a cost of \$632 million and \$157 million, respectively. Our repurchases during the six months ended June 30, 2022 consisted of 4.6 million shares at a cost of \$582 million under our Rule 10b5-1 trading plan and 0.4 million shares at a cost of \$50 million on the open market during an open trading period. All shares repurchased during the three months ended June 30, 2022 were done under our Rule 10b5-1 trading plan. As of June 30, 2023, the remaining balance of Board approved funds for future repurchases was \$2.5 billion. In connection with our pending acquisition of Black Knight, on May 4, 2022 we terminated our Rule 10b5-1 trading plan and suspended share repurchases.

Dividends

During the six months ended June 30, 2023 and 2022, we declared and paid cash dividends per share of \$0.84 and \$0.76, respectively, for an aggregate payout of \$472 million and \$427 million, respectively. During the three months ended June 30, 2023 and 2022, we declared dividends per share of \$0.42 and \$0.38, respectively for an aggregate payout of \$236 million and \$213 million, respectively. The declaration of dividends is subject to the discretion of our Board. Our Board has adopted a quarterly dividend declaration policy providing that the declaration of any dividends will be determined quarterly by the Board or the Audit Committee, taking into account such factors as our evolving business model, prevailing business conditions, our financial results and capital requirements and other considerations which our Board deems relevant, without a predetermined annual net income payout ratio.

Accumulated Other Comprehensive Income/(Loss)

The following tables present changes in the accumulated balances for each component of other comprehensive income/ (loss) (in millions):

Changes in Accumulated Other Comprehensive Income/(Loss) by Component

	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2022	\$ (278)	\$ 2	\$ (55)	\$ (331)
Other comprehensive income	47	—	—	47
Income tax benefit/(expense)	—	—	—	—
Net current period other comprehensive income	47	—	—	47
Balance, as of June 30, 2023	<u>\$ (231)</u>	<u>\$ 2</u>	<u>\$ (55)</u>	<u>\$ (284)</u>

Changes in Accumulated Other Comprehensive Income/(Loss) by Component

	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of March 31, 2023	\$ (262)	\$ 2	\$ (55)	\$ (315)
Other comprehensive income	31	—	—	31
Income tax benefit/(expense)	—	—	—	—
Net current period other comprehensive income	31	—	—	31
Balance, as of June 30, 2023	<u>\$ (231)</u>	<u>\$ 2</u>	<u>\$ (55)</u>	<u>\$ (284)</u>

Changes in Accumulated Other Comprehensive Income/(Loss) by Component

	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2021	\$ (150)	\$ 2	\$ (48)	\$ (196)
Other comprehensive loss	(110)	—	—	(110)
Income tax benefit/(expense)	1	—	—	1
Net current period other comprehensive loss	(109)	—	—	(109)
Balance, as of June 30, 2022	<u>\$ (259)</u>	<u>\$ 2</u>	<u>\$ (48)</u>	<u>\$ (305)</u>

Changes in Accumulated Other Comprehensive Income/(Loss) by Component

	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of March 31, 2022	\$ (175)	\$ 2	\$ (48)	\$ (221)
Other comprehensive loss	(85)	—	—	(85)
Income tax benefit/(expense)	1	—	—	1
Net current period other comprehensive loss	(84)	—	—	(84)
Balance, as of June 30, 2022	<u>\$ (259)</u>	<u>\$ 2</u>	<u>\$ (48)</u>	<u>\$ (305)</u>

11. Income Taxes

Our effective tax rate was 12% and 22% during the six months ended June 30, 2023 and 2022, respectively, and 4% and 23% during the three months ended June 30, 2023 and 2022, respectively. The effective tax rates for the six months and three months ended June 30, 2023 were lower than the effective tax rates for the comparable periods in 2022 primarily due to the tax benefits resulting from favorable audit settlements for prior years, deferred and current tax benefits from state apportionment changes and the application of the high-tax exception to the Global Intangible Low-Taxed Income, or GILTI, in the current period, partially offset by the impact of the U.K. corporate income tax increase from 19% to 25% effective April 1, 2023.

In conjunction with the increase in the U.K. corporate income tax rate, we intend to elect the GILTI high-tax exception election in 2023. During the three months ended June 30, 2023, our tax provision includes the impacts of this election, which resulted in a reduction in our GILTI inclusions. Our unrecognized tax benefit as of June 30, 2023 was \$240 million, a \$7 million net decrease from the \$247 million as of December 31, 2022. The net decrease includes a \$40 million

reduction as a result of audit settlements, a \$20 million increase related to current year positions, a \$31 million increase related to prior year positions, and an \$18 million reduction related to prior year positions.

In August 2022, the Inflation Reduction Act of 2022, or IRA, was signed into law. The IRA introduced a 15% corporation minimum tax, or CAMT, on adjusted financial statement income for corporations with profits in excess of \$1 billion, effective for tax years after December 31, 2022. Based on the current guidance provided by the Internal Revenue Service and Treasury, the implementation of the CAMT does not have a material impact to our financial statements as of June 30, 2023.

The IRA also includes a share buyback excise tax of 1% on share repurchases, which will apply to net share repurchases after December 31, 2022. During the six months ended June 30, 2023, we did not repurchase any shares, therefore, we were not subject to any excise tax. The newly imposed excise tax on share repurchases is not considered an income tax. Any excise tax, as a result of future share repurchases, will be considered part of the cost of the shares repurchased and reflected in the equity section of our consolidated financial statements.

12. Clearing Operations

We operate six clearing houses, each of which acts as a central counterparty that becomes the buyer to every seller and the seller to every buyer for its clearing members or participants, or Members. Through this central counterparty function, the clearing houses provide financial security for each transaction for the duration of the position by limiting counterparty credit risk.

Our clearing houses are responsible for providing clearing services to each of our futures exchanges, and in some cases to third-party execution venues, and are as follows, referred to herein collectively as "the ICE Clearing Houses":

Clearing House	Products Cleared	Exchange where Executed	Location
ICE Clear Europe	Energy, agricultural, interest rates and equity index futures and options contracts and OTC European CDS instruments	ICE Futures Europe, ICE Futures U.S., ICE Endex, ICE Futures Abu Dhabi and third-party venues	U.K.
ICE Clear U.S.	Agricultural, metals, foreign exchange, or FX, interest rate, equity index and digital asset futures and/or options contracts	ICE Futures U.S.	U.S.
ICE Clear Credit	OTC North American, European, Asian-Pacific and Emerging Market CDS instruments	Creditex and third-party venues	U.S.
ICE Clear Netherlands	Derivatives on equities and equity indices traded on regulated markets	ICE Endex	The Netherlands
ICE Clear Singapore	Energy, metals and financial futures products and digital asset futures contracts	ICE Futures Singapore	Singapore
ICE NGX	Physical North American natural gas and electricity	ICE NGX	Canada

In 2022, we announced our decision to cease our CDS clearing service at ICE Clear Europe, our clearing house in the U.K., and thereafter our sole CDS clearing offering will be at our ICE Clear Credit clearing house in the U.S. This is expected to be completed in late 2023.

Original and Variation Margin

Each of the ICE Clearing Houses generally requires all Members to deposit collateral in cash or certain pledged assets. The collateral deposits are known as "original margin." In addition, the ICE Clearing Houses may make intraday original margin calls in circumstances where market conditions require additional protection. The daily profits and losses to and from the ICE Clearing Houses due to the marking-to-market of open contracts is known as "variation margin." The ICE Clearing Houses mark all outstanding contracts to market, and, with the exception of ICE NGX's physical natural gas and physical power products discussed separately below, pay and collect variation margin, at least once daily.

The amounts that Members are required to maintain are determined by proprietary risk models established by each ICE Clearing House and reviewed by the relevant regulators, independent model validators, risk committees and the boards of directors of the respective ICE Clearing House. The amounts required may fluctuate over time. Each of the ICE Clearing Houses is a separate legal entity and is not subject to the liabilities of the others, or the obligations of Members of the other ICE Clearing Houses.

Should a particular Member fail to deposit its original margin or fail to make a variation margin payment, when and as required, the relevant ICE Clearing House may liquidate or hedge the defaulting Member's open positions and use their original margin and guaranty fund deposits to pay any amount owed. In the event that the defaulting Member's deposits are not sufficient to pay the amount owed in full, the ICE Clearing Houses will first use their respective contributions to the

guaranty fund, often referred to as Skin In The Game, or SITG, to pay any remaining amount owed. In the event that the SITG is not sufficient, the ICE Clearing Houses may utilize the respective guaranty fund deposits and default insurance, or collect limited additional funds from their respective non-defaulting Members on a pro-rata basis, to pay any remaining amount owed.

As of June 30, 2023 and December 31, 2022, the ICE Clearing Houses had received or had been pledged \$194.0 billion and \$273.3 billion, respectively, in cash and non-cash collateral in original margin and guaranty fund deposits to cover price movements of underlying contracts for both periods.

Guaranty Funds and ICE Contribution

As described above, mechanisms have been created, called guaranty funds, to provide partial protection in the event of a Member default. With the exception of ICE NGX, each of the ICE Clearing Houses requires that each Member make deposits into a guaranty fund.

In addition, we have contributed our own capital that could be used if a defaulting Member’s original margin and guaranty fund deposits are insufficient. Included in the total contribution to ICE Clear U.S., as of June 30, 2023, is \$15 million from Bakkt, solely applicable to any losses associated with a default in Bitcoin contracts and other digital assets that ICE Clear U.S. may clear in the future. Such amounts are recorded as long-term restricted cash and cash equivalents or long-term restricted investments in our balance sheets and are as follows (in millions):

Clearing House	ICE Portion of Guaranty Fund Contribution		Default insurance	
	As of June 30, 2023	As of December 31, 2022	As of June 30, 2023	As of December 31, 2022
ICE Clear Europe	\$247	\$247	\$100	\$100
ICE Clear U.S.	90	90	25	25
ICE Clear Credit	50	50	75	75
ICE Clear Netherlands	2	2	N/A	N/A
ICE Clear Singapore	1	1	N/A	N/A
ICE NGX	15	15	200	200
Total	\$405	\$405	\$400	\$400

We also maintain default insurance as an additional layer of clearing member default protection. The default insurance was renewed in September 2022 and has a three-year term for the following clearing houses in the following amounts: ICE Clear Europe - \$100 million; ICE Clear U.S. - \$25 million; and ICE Clear Credit - \$75 million. The default insurance layer resides after and in addition to the ICE Clear Europe, ICE Clear U.S. and ICE Clear Credit SITG contributions and before the guaranty fund contributions of the non-defaulting Members.

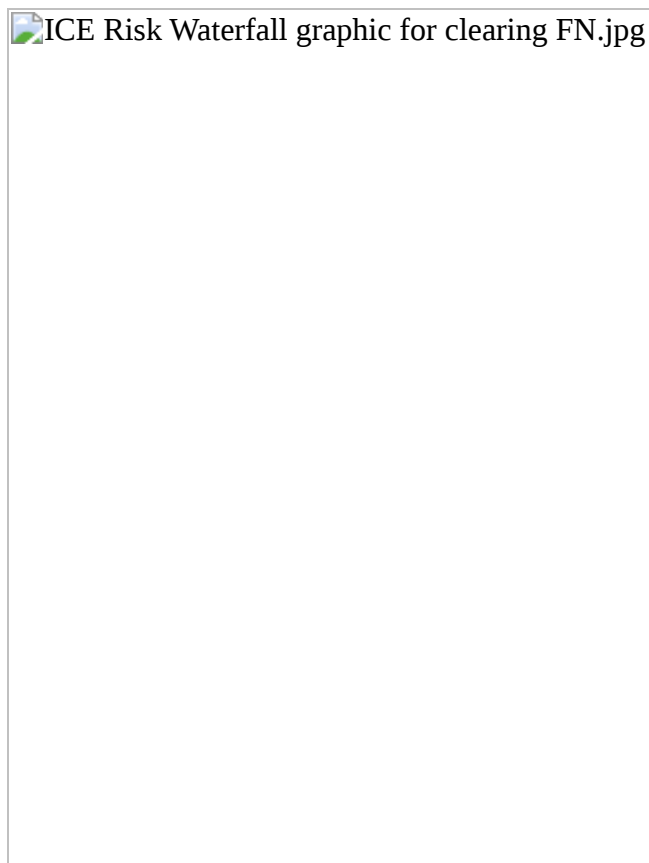
Similar to SITG, the default insurance layer is not intended to replace or reduce the position risk-based amount of the guaranty fund. As a result, the default insurance layer is not a factor that is included in the calculation of the Members' guaranty fund contribution requirement. Instead, it serves as an additional, distinct, and separate default resource that should serve to further protect the non-defaulting Members’ guaranty fund contributions from being mutualized in the event of a default.

As of June 30, 2023, ICE NGX maintained a guaranty fund of \$215 million, comprised of \$15 million in cash and a \$200 million letter of credit backed by a default insurance policy of the same amount, discussed below.

Below is a depiction of our Default Waterfall which summarizes the lines of defense and layers of protection we maintain for our mutualized clearing houses.

ICE Clearing House Default Waterfall

 ICE Risk Waterfall graphic for clearing FN.jpg



Cash and Invested Margin Deposits

We have recorded cash and invested margin and guaranty fund deposits and amounts due in our balance sheets as current assets with corresponding current liabilities to the Members. As of June 30, 2023, our cash and invested margin deposits were as follows (in millions):

	ICE Clear Europe ⁽¹⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 45,464	\$ 29,056	\$ 4,968	\$ —	\$ 5	\$ 79,493
Unsettled variation margin, net	—	—	—	693	—	693
Guaranty fund	3,886	3,197	646	—	4	7,733
Delivery contracts receivable/payable, net	—	—	—	410	—	410
Total	\$ 49,350	\$ 32,253	\$ 5,614	\$ 1,103	\$ 9	\$ 88,329

As of December 31, 2022, our cash and invested margin deposits were as follows (in millions):

	ICE Clear Europe ⁽²⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 101,243	\$ 31,277	\$ 4,141	\$ —	\$ 5	\$ 136,666
Unsettled variation margin, net	—	—	—	749	—	749
Guaranty fund	4,162	3,177	597	—	4	7,940
Delivery contracts receivable/payable, net	—	—	—	2,017	—	2,017
Total	\$ 105,405	\$ 34,454	\$ 4,738	\$ 2,766	\$ 9	\$ 147,372

⁽¹⁾ \$44.7 billion and \$4.6 billion is related to futures/options and CDS, respectively.

⁽²⁾ \$97.6 billion and \$7.8 billion is related to futures/options and CDS, respectively.

Our cash and invested margin and guaranty fund deposits are maintained in accounts with national banks and highly-rated financial institutions or secured through direct investments, primarily in U.S. Treasury and other highly-rated foreign government securities, or reverse repurchase agreements with primarily overnight maturities. We primarily use Level 1

inputs when evaluating the fair value of the non-cash equivalent direct investments, as highly-rated government securities are quoted in active markets. The carrying value of these deposits is deemed to approximate fair value.

To provide a tool to address the liquidity needs of our clearing houses and manage the liquidation of margin and guaranty fund deposits held in the form of cash and high quality sovereign debt, ICE Clear Europe, ICE Clear Credit and ICE Clear U.S. have entered into Committed Repurchase Agreement Facilities, or Committed Repo. Additionally, ICE Clear Credit and ICE Clear Netherlands have entered into Committed FX Facilities to support these liquidity needs. As of June 30, 2023, the following facilities were in place:

- **ICE Clear Europe:** \$1.0 billion in Committed Repo to finance U.S. dollar, euro and pound sterling deposits.
- **ICE Clear Credit:** \$300 million in Committed Repo (U.S. dollar based) to finance U.S. dollar denominated sovereign debt and euro deposits, €250 million in Committed Repo (euro based) to finance euro and U.S. dollar denominated sovereign debt deposits, and €1.9 billion in Committed FX Facilities to finance euro payment obligations.
- **ICE Clear U.S.:** \$250 million in Committed Repo to finance U.S. dollar denominated sovereign debt deposits.
- **ICE Clear Netherlands:** €10 million in Committed FX Facilities to finance euro payment obligations.

Details of our deposits are as follows (in millions):

Cash and Cash Equivalent Margin Deposits and Guaranty Funds			
Clearing House	Investment Type	As of June 30, 2023	As of December 31, 2022
ICE Clear Europe	National bank account ⁽¹⁾	\$ 5,932	\$ 17,390
ICE Clear Europe	Reverse repo	41,300	65,352
ICE Clear Europe	Sovereign debt	1,791	19,894
ICE Clear Europe	Demand deposits	16	153
ICE Clear Credit	National bank account	24,910	27,145
ICE Clear Credit	Reverse repo	4,310	3,916
ICE Clear Credit	Demand deposits	3,034	3,393
ICE Clear U.S.	Reverse repo	5,465	4,266
ICE Clear U.S.	Sovereign Debt	149	472
Other ICE Clearing Houses	Demand deposits	10	9
Total cash and cash equivalent margin deposits and guaranty funds		\$ 86,917	\$ 141,990

Invested Deposits, Delivery Contracts Receivable and Unsettled Variation Margin			
Clearing House	Investment Type	As of June 30, 2023	As of December 31, 2022
ICE NGX	Unsettled variation margin and delivery contracts receivable/payable	1,103	2,766
ICE Clear Europe	Invested deposits - sovereign debt	309	2,616
Total invested deposits, delivery contracts receivable and unsettled variation margin		\$ 1,412	\$ 5,382

⁽¹⁾ As of June 30, 2023, ICE Clear Europe held €58 million (\$63 million based on the euro/U.S. dollar exchange rate of 1.0911 as of June 30, 2023) at the European Central Bank, or ECB, £4.6 billion (\$5.9 billion based on the pound sterling/U.S. dollar exchange rate of 1.2697 as of June 30, 2023) at the Bank of England, or BOE, and €10 million (\$11 million based on the above exchange rate) at the BOE. As of December 31, 2022, ICE Clear Europe held €11.7 billion (\$12.5 billion based on the euro/U.S. dollar exchange rate of 1.0704 as of December 31, 2022) at ECB, £4.0 billion (\$4.9 billion based on the pound sterling/U.S. dollar exchange rate of 1.2093 as of December 31, 2022) at the BOE and €10 million (\$11 million based on the above exchange rate) at the BOE.

Other Deposits

Non-cash original margin and guaranty fund deposits are not reflected in the accompanying consolidated balance sheets as the risks and rewards of these assets remain with the clearing members unless the clearing houses have sold or re-pledged the assets or in the event of a clearing member default, where the clearing member is no longer entitled to redeem the assets. Any income, gain or loss accrues to the clearing members.

In addition to the cash and invested deposits above, the ICE Clearing Houses have also received other assets from Members, which include government obligations, emissions allowances, and may include other non-cash collateral such

as letters of credit at ICE NGX to mitigate credit risk. For certain deposits, we may impose discount or “haircut” rates to ensure adequate collateral if market values fluctuate. The value-related risks and rewards of these assets remain with the Members. Any gain or loss accrues to the Member. The ICE Clearing Houses do not, in the ordinary course, rehypothecate or re-pledge these assets. These pledged assets are not reflected in our balance sheets, and are as follows (in millions):

As of June 30, 2023					
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 53,228	\$ 31,705	\$ 12,920	\$ —	\$ 97,853
Letters of credit and other	237	—	—	4,326	4,563
ICE NGX cash deposits	—	—	—	1,349	1,349
Total	<u>\$ 53,465</u>	<u>\$ 31,705</u>	<u>\$ 12,920</u>	<u>\$ 5,675</u>	<u>\$ 103,765</u>
Guaranty fund:					
Government securities at face value	<u>\$ 765</u>	<u>\$ 846</u>	<u>\$ 289</u>	<u>\$ —</u>	<u>\$ 1,900</u>

As of December 31, 2022					
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 74,964	\$ 26,601	\$ 14,855	\$ —	\$ 116,420
Letters of credit	—	—	—	5,434	5,434
ICE NGX cash deposits	—	—	—	2,357	2,357
Total	<u>\$ 74,964</u>	<u>\$ 26,601</u>	<u>\$ 14,855</u>	<u>\$ 7,791</u>	<u>\$ 124,211</u>
Guaranty fund:					
Government securities at face value	<u>\$ 641</u>	<u>\$ 805</u>	<u>\$ 269</u>	<u>\$ —</u>	<u>\$ 1,715</u>

The ICE Clearing Houses invest cash margin deposits, including both the guaranty fund and original margin deposits on behalf of clearing members, primarily for purposes of safeguarding customer funds. Interest earned on cash margin investments is provided back to clearing members net of certain costs and administrative fees charged and retained by ICE. The ICE Clearing Houses also charge fees for clearing members pledging non-cash margin in lieu of cash margin, these fees are fully retained by ICE. The net interest income on cash margin and fees charged for non-cash margin retained by the ICE Clearing Houses is recorded in our Exchanges segment as OTC and other revenues and in our Fixed Income and Data Services segment as CDS clearing revenues in our consolidated statement of income. We recognized a combined \$191 million and \$139 million as our revenues during the six months ended June 30, 2023 and 2022, respectively, and \$91 million and \$81 million as our revenues during the three months ended June 30, 2023 and 2022, respectively.

ICE NGX

ICE NGX owns a clearing house which primarily administers the physical delivery of energy trading contracts. ICE NGX is the central counterparty to Members on opposite sides of its physically-settled contracts, and the balance related to delivered but unpaid contracts is recorded as a delivery contract net receivable, with an offsetting delivery contract net payable in our balance sheets. Unsettled variation margin equal to the fair value of open contracts is recorded as of each balance sheet date. There is no impact on our consolidated statements of income as an equal amount is recognized as both an asset and a liability. ICE NGX marks all its outstanding physical natural gas and physical power contracts to market daily, but only collects variation margin when a Member's open position falls outside a specified percentage of its pledged collateral. Due to the highly liquid nature and the short period of time to maturity, the fair values of our delivery contract net payable and net receivable are determined to approximate carrying value.

ICE NGX requires Members to maintain cash or letters of credit to serve as collateral in the event of default. The cash is maintained in a segregated bank account for the benefit of the Member, and remains the property of the Member, therefore, it is not included in our balance sheets. ICE NGX maintains a committed daylight-overnight liquidity facility in the amount of \$100 million with an additional \$200 million uncommitted with a third-party Canadian chartered bank which provides liquidity in the event of a settlement shortfall, subject to certain conditions.

As of June 30, 2023, ICE NGX maintains a guaranty fund of \$215 million funded by a \$200 million letter of credit issued by a major Canadian chartered bank, and backed by default insurance underwritten by Export Development Canada, or

EDC, a Crown corporation operated at arm's length from the Canadian government, plus \$15 million held as restricted cash to fund the first loss amount that ICE NGX is responsible for under the default insurance policy. In the event of a participant default where the Member's collateral is depleted, the shortfall would be covered by a draw down on the letter of credit following which ICE NGX would file a claim under the default insurance to recover additional losses up to \$200 million beyond the \$15 million first-loss amount that ICE NGX is responsible for under the default insurance policy.

Clearing House Exposure

The net notional value of unsettled contracts was \$2.4 trillion as of June 30, 2023. Each ICE Clearing House bears financial counterparty credit risk and provides a central counterparty guarantee, or performance guarantee, to its Members. To reduce their exposure, the ICE Clearing Houses have a risk management program with both initial and ongoing membership standards. Excluding the effects of original and variation margin, guaranty fund and collateral requirements and default insurance, the ICE Clearing Houses' maximum estimated exposure for this guarantee is \$161.8 billion as of June 30, 2023, which represents the maximum estimated value by the ICE Clearing Houses of a hypothetical one-day movement in pricing of the underlying unsettled contracts. This value was determined using proprietary risk management software that simulates gains and losses based on historical market prices, volatility and other factors present at that point in time for those particular unsettled contracts. Future market price volatility could result in the exposure being significantly different than this amount.

13. Legal Proceedings

In the ordinary course of our business, from time to time we are subject to legal proceedings, lawsuits, government investigations and other claims with respect to a variety of matters. In addition, we are subject to periodic reviews, inspections, examinations and investigations by regulators in the U.S. and other jurisdictions, any of which may result in claims, legal proceedings, assessments, fines, penalties, restrictions on our business or other sanctions. We record estimated expenses and reserves for legal or regulatory matters or other claims when these matters present loss contingencies that are probable and the related amount is reasonably estimable. Any such accruals may be adjusted as circumstances change. Assessments of losses are inherently subjective and involve unpredictable factors. While the outcome of legal and regulatory matters is inherently difficult to predict and/or the range of loss often cannot be reasonably estimable, we do not believe that the liabilities, other than the potential \$725 million termination fee payable to Black Knight and our accruals related to potential regulatory settlements of \$11 million, which may ultimately result from the resolution of the various legal and regulatory matters that arise in the ordinary course of our business, including the matter described below and those described in Note 16 to the consolidated financial statements in Part II, Item 8 of our 2022 Form 10-K, are likely to have a material adverse effect on our consolidated financial condition, results of operations, or liquidity. It is possible, however, that future results of operations for any particular quarterly or annual period could be materially and adversely affected by any developments relating to these legal and regulatory matters. A range of possible losses related to certain cases cannot be reasonably estimated at this time, except as otherwise disclosed below and in Note 16 to the consolidated financial statements in Part II, Item 8 of our 2022 Form 10-K. Individual matter disclosures in this Form 10-Q are limited to new significant matters or significant updates on existing matters since the 2022 Form 10-K.

Black Knight Transaction Litigation

On March 9, 2023, the Federal Trade Commission, or the FTC, filed an administrative complaint alleging that the proposed transaction between ICE and Black Knight, if consummated, would be an unfair method of competition in violation of Section 5 of the Federal Trade Commission Act, and that it would substantially lessen competition, or tend to create a monopoly, in violation of Section 7 of the Clayton Act. The complaint seeks a variety of injunctive relief, including, among other things, a prohibition on the completion of the transaction without the FTC's consent and, if the transaction is completed, a divestiture or reconstitution of assets in a manner that restores such separate and independent businesses as the parties had operated prior to the completion of the transaction. On April 10, 2023, the FTC filed a complaint in the United States District Court for the Northern District of California for a temporary restraining order and preliminary injunction enjoining the completion of the transaction. On April 21, 2023, the court entered a temporary restraining order enjoining the completion of the transaction until the court rules on the FTC's motion for a preliminary injunction. In their answers to the administrative and court complaints, filed on March 20, 2023 and April 25, 2023, respectively, ICE and Black Knight denied the FTC's substantive allegations; asserted numerous affirmative defenses; described the pro-competitive aspects and significant lender, servicer, investor, vendor and consumer benefits relating to this transaction; and denied that the combination of their respective businesses would violate any laws. Additionally, the answers to the court complaint contained counterclaims by ICE and Black Knight against the FTC seeking declaratory relief that the FTC's administrative process is unconstitutional and should be enjoined. On July 17, 2023, the district court entered an order granting a joint motion by the parties to continue the evidentiary hearing on the FTC's motion for a preliminary injunction, and rescheduled the hearing to August 14, 2023. In continuing the hearing, the court noted that the parties are discussing a potential resolution of the matter resulting from the announcement of the planned divestiture of the Optimal

Blue business (see Note 3) and the FTC’s analysis of the implications of the divestiture for this case and the administrative complaint. On July 19, 2023, FTC counsel filed an unopposed motion with the FTC to withdraw the administrative complaint from adjudication; on July 25, 2023, the FTC granted this motion. We are vigorously defending against the FTC’s court complaint.

For further information on our legal and regulatory matters, please see Note 16 to the consolidated financial statements in Part II, Item 8 of our 2022 Form 10-K.

14. Fair Value Measurements

Fair value is the price that would be received from selling an asset or paid to transfer a liability (an exit price) in the principal or most advantageous market for the asset or liability in an orderly transaction between market participants at the measurement date. Our financial instruments consist primarily of certain short-term and long-term assets and liabilities, customer accounts receivable, margin deposits and guaranty funds, equity and equity method investments, and short-term and long-term debt.

The fair value of our financial instruments is measured based on a three-level hierarchy:

- **Level 1 inputs** — quoted prices for identical assets or liabilities in active markets.
- **Level 2 inputs** — observable inputs other than Level 1 inputs such as quoted prices for similar assets and liabilities in active markets or inputs other than quoted prices that are directly observable.
- **Level 3 inputs** — unobservable inputs supported by little or no market activity and that are significant to the fair value of the assets or liabilities.

Financial assets and liabilities recorded or disclosed at fair value in the accompanying consolidated balance sheets as of June 30, 2023 and December 31, 2022 were classified in their entirety based on the lowest level of input that is significant to the asset or liability’s fair value measurement.

Our restricted short-term and long-term investments represent restricted regulatory funds and SITG funds, respectively, at ICE Clear Europe, invested in treasury securities with maturities of greater than 90 days.

Our mutual funds are equity and fixed income mutual funds held for the purpose of providing future payments for our supplemental executive savings plan and the supplemental executive retirement plan. These mutual funds are classified as equity investments and measured at fair value using Level 1 inputs with adjustments recorded in net income.

Excluding our equity investments without a readily determinable fair value, all other financial instruments are determined to approximate carrying value due to the short period of time to their maturities.

We did not use Level 3 inputs to determine the fair value of assets or liabilities measured at fair value on a recurring basis as of June 30, 2023 or December 31, 2022.

We measure certain assets, such as intangible assets and equity method investments, at fair value on a non-recurring basis. These assets are recognized at fair value if they are deemed to be impaired. As of December 31, 2022, certain equity method investments were measured at fair value on a non-recurring basis. As of June 30, 2023, none of our intangible assets or equity method investments were required to be recorded at fair value since no impairments were recorded.

We measure certain equity investments at fair value on a non-recurring basis using our policy election under ASU 2016-01. During the six months ended June 30, 2023, we evaluated these investments and determined that no fair value adjustments were required under our accounting policy election related to these investments.

See Note 12 for the fair value considerations related to our margin deposits, guaranty funds and delivery contracts receivable.

The table below displays the fair value of our debt as of June 30, 2023. The fair values of our fixed rate notes were estimated using quoted market prices for these instruments. The fair value of other short-term debt approximates par value since the interest rates on this short-term debt approximate market rates as of June 30, 2023.

As of June 30, 2023

(in millions)

Debt:	Carrying Amount	Fair value
3.65% Senior Notes due May 23, 2025	\$ 1,245	\$ 1,226
3.75% Senior Notes due December 1, 2025	1,248	1,214
4.00% Senior Notes due September 15, 2027	1,488	1,463
3.10% Senior Notes due September 15, 2027	498	468
3.75% Senior Notes due September 21, 2028	595	571
4.35% Senior Notes due June 15, 2029	1,240	1,225
2.10% Senior Notes due June 15, 2030	1,237	1,046
1.85% Senior Notes due September 15, 2032	1,485	1,157
4.60% Senior Notes due March 15, 2033	1,488	1,455
2.65% Senior Notes due September 15, 2040	1,232	892
4.25% Senior Notes due September 21, 2048	1,232	1,091
3.00% Senior Notes due June 15, 2050	1,221	871
4.95% Senior Notes due June 15, 2052	1,465	1,435
3.00% Senior Notes due September 15, 2060	1,471	985
5.20% Senior Notes due June 15, 2062	983	996
Total debt	\$ 18,128	\$ 16,095

15. Segment Reporting

Our business is conducted through three reportable business segments:

- **Exchanges:** We operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities;
- **Fixed Income and Data Services:** We provide fixed income pricing, reference data, indices, analytics and execution services as well as global CDS clearing and multi-asset class data delivery solutions; and
- **Mortgage Technology:** We provide a technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

While revenues are recorded specifically in the segment in which they are earned or to which they relate, a significant portion of our operating expenses are not solely related to a specific segment because the expenses serve functions that are necessary for the operation of more than one segment. We directly allocate expenses when reasonably possible to do so. Otherwise, we use a pro-rata revenue approach as the allocation method for the expenses that do not relate solely to one segment and serve functions that are necessary for the operation of all segments.

Our chief operating decision maker does not review total assets or statements of income below operating income by segments; therefore, such information is not presented below. Our three segments do not engage in intersegment transactions.

Beginning in the first quarter of 2023, closing solutions revenues within our Mortgage Technology segment now include membership dues that were previously included in other revenues. We believe this is a more accurate reflection of the nature of these revenues. The impact of this change was not material, and the prior year periods have been adjusted for comparability.

Financial data for our business segments is as follows for the six months and three months ended June 30, 2023 and 2022 (in millions):

	Six Months Ended June 30, 2023			
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 700	\$ —	\$ —	\$ 700
Agricultural and metals futures and options	147	—	—	147
Financial futures and options	232	—	—	232
Cash equities and equity options	1,215	—	—	1,215
OTC and other	205	—	—	205
Data and connectivity services	463	—	—	463
Listings	252	—	—	252
Fixed income execution	—	60	—	60
CDS clearing	—	185	—	185
Fixed income data and analytics	—	553	—	553
Other data and network services	—	311	—	311
Origination technology	—	—	337	337
Closing solutions	—	—	87	87
Data and analytics	—	—	45	45
Other	—	—	16	16
Revenues	3,214	1,109	485	4,808
Transaction-based expenses	1,024	—	—	1,024
Revenues, less transaction-based expenses	2,190	1,109	485	3,784
Operating expenses	631	699	530	1,860
Operating income/(loss)	\$ 1,559	\$ 410	\$ (45)	\$ 1,924
	Three Months Ended June 30, 2023			
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 355	\$ —	\$ —	\$ 355
Agricultural and metals futures and options	77	—	—	77
Financial futures and options	104	—	—	104
Cash equities and equity options	544	—	—	544
OTC and other	104	—	—	104
Data and connectivity services	231	—	—	231
Listings	126	—	—	126
Fixed income execution	—	28	—	28
CDS clearing	—	84	—	84
Fixed income data and analytics	—	277	—	277
Other data and network services	—	157	—	157
Origination technology	—	—	170	170
Closing solutions	—	—	47	47
Data and analytics	—	—	24	24
Other	—	—	8	8
Revenues	1,541	546	249	2,336
Transaction-based expenses	448	—	—	448
Revenues, less transaction-based expenses	1,093	546	249	1,888
Operating expenses	311	356	266	933
Operating income/(loss)	\$ 782	\$ 190	\$ (17)	\$ 955

	Six Months Ended June 30, 2022			
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 618	\$ —	\$ —	\$ 618
Agricultural and metals futures and options	122	—	—	122
Financial futures and options	253	—	—	253
Cash equities and equity options	1,357	—	—	1,357
OTC and other	205	—	—	205
Data and connectivity services	432	—	—	432
Listings	260	—	—	260
Fixed income execution	—	40	—	40
CDS clearing	—	138	—	138
Fixed income data and analytics	—	551	—	551
Other data and network services	—	292	—	292
Origination technology	—	—	399	399
Closing solutions	—	—	138	138
Data and analytics	—	—	44	44
Other	—	—	23	23
Revenues	3,247	1,021	604	4,872
Transaction-based expenses	1,159	—	—	1,159
Revenues, less transaction-based expenses	2,088	1,021	604	3,713
Operating expenses	603	692	557	1,852
Operating income	\$ 1,485	\$ 329	\$ 47	\$ 1,861

	Three Months Ended June 30, 2022			
	Exchanges	Fixed Income and Data Services	Mortgage Technology	Consolidated
Revenues:				
Energy futures and options	\$ 265	\$ —	\$ —	\$ 265
Agricultural and metals futures and options	61	—	—	61
Financial futures and options	123	—	—	123
Cash equities and equity options	698	—	—	698
OTC and other	108	—	—	108
Data and connectivity services	218	—	—	218
Listings	131	—	—	131
Fixed income execution	—	25	—	25
CDS clearing	—	66	—	66
Fixed income data and analytics	—	274	—	274
Other data and network services	—	147	—	147
Origination technology	—	—	196	196
Closing solutions	—	—	66	66
Data and analytics	—	—	24	24
Other	—	—	11	11
Revenues	1,604	512	297	2,413
Transaction-based expenses	599	—	—	599
Revenues, less transaction-based expenses	1,005	512	297	1,814
Operating expenses	304	338	303	945
Operating income/(loss)	\$ 701	\$ 174	\$ (6)	\$ 869

Revenue from one member of the Exchanges segment comprised \$254 million, or 12%, and \$130 million, or 12%, of our Exchange revenues, less transaction-based expenses for the six months and three months ended June 30, 2023, respectively. Revenue from one member of the Exchanges segment comprised \$213 million, or 10%, of our Exchange

revenues, less transaction-based expenses for the six months ended June 30, 2022 and no customers or clearing members accounted for more than 10% of our Exchange revenues, less transaction-based expenses during the three months ended June 30, 2022. Clearing members are primarily intermediaries and represent a broad range of principal trading firms. If a clearing member ceased its operations, we believe that the trading firms would continue to conduct transactions and would clear those transactions through another clearing member firm. No additional customers or clearing members accounted for more than 10% of our segment revenues or consolidated revenues during the six months or three months ended June 30, 2023 or 2022.

16. Earnings Per Common Share

The following is a reconciliation of the numerators and denominators of the basic and diluted earnings per common share computations for the six months and three months ended June 30, 2023 and 2022 (in millions, except per share amounts):

	Six Months Ended June 30,		Three Months Ended June 30,	
	2023	2022	2023	2022
Basic:				
Net income attributable to Intercontinental Exchange, Inc.	\$ 1,454	\$ 1,212	\$ 799	\$ 555
Weighted average common shares outstanding	560	560	560	558
Basic earnings per common share	\$ 2.60	\$ 2.17	\$ 1.43	\$ 0.99
Diluted:				
Weighted average common shares outstanding	560	560	560	558
Effect of dilutive securities - stock options and restricted stock	1	2	1	2
Diluted weighted average common shares outstanding	561	562	561	560
Diluted earnings per common share	\$ 2.59	\$ 2.16	\$ 1.42	\$ 0.99

Basic earnings per common share is calculated using the weighted average common shares outstanding during the period. Common equivalent shares from stock options and restricted stock awards, calculated using the treasury stock method, are included in the diluted per share calculations unless the effect of their inclusion would be antidilutive. During both the six months ended June 30, 2023 and 2022, 1 million outstanding stock options and restricted stock awards were not included in the computation of diluted earnings per common share, because to do so would have had an antidilutive effect.

17. Subsequent Events

We have evaluated subsequent events, and determined that no events or transactions, other than those already disclosed in this Quarterly Report, met the definition of a subsequent event for purposes of recognition or disclosure in the accompanying consolidated financial statements.

ITEM 2. MANAGEMENT’S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

In this Quarterly Report on Form 10-Q, or this Quarterly Report, and unless otherwise indicated, the terms “Intercontinental Exchange,” “ICE,” “we,” “us,” “our,” “our company” and “our business” refer to Intercontinental Exchange, Inc., together with its consolidated subsidiaries. All references to “options” or “options contracts” in the context of our futures products refer to options on futures contracts. Solely for convenience, references in this Quarterly Report to any trademarks, service marks and trade names owned by ICE are listed without the ®, ™ and © symbols, but we will assert, to the fullest extent under applicable law, our rights to these trademarks, service marks and trade names.

We also include references to third-party trademarks, trade names and service marks in this Quarterly Report. Except as otherwise expressly noted, our use or display of any such trademarks, trade names or service marks is not an endorsement or sponsorship and does not indicate any relationship between us and the parties that own such marks and names.

The following discussion should be read in conjunction with our consolidated financial statements and related notes included elsewhere in this Quarterly Report. Due to rounding, figures in tables may not sum exactly.

Forward-Looking Statements

This Quarterly Report, including the sections entitled “Notes to Consolidated Financial Statements,” “Legal Proceedings” and “Management’s Discussion and Analysis of Financial Condition and Results of Operations,” contains “forward-looking statements” as defined in the Private Securities Litigation Reform Act of 1995. Any statements contained herein that are not statements of historical fact may be forward-looking statements.

These forward-looking statements relate to future events or our future financial performance and are based on our present beliefs and assumptions as well as the information currently available to us. They involve known and unknown risks, uncertainties and other factors that may cause our results, levels of activity, performance, cash flows, financial position or achievements to differ materially from those expressed or implied by these statements.

Forward-looking statements may be introduced by or contain terminology such as “may,” “will,” “should,” “could,” “would,” “targets,” “goal,” “expect,” “intend,” “plan,” “anticipate,” “believe,” “estimate,” “predict,” “potential,” “continue,” or the antonyms of these terms or other comparable terminology. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, cash flows, financial position or achievements. Accordingly, we caution you not to place undue reliance on any forward-looking statements we may make.

Factors that may affect our performance and the accuracy of any forward-looking statements include, but are not limited to, those listed below:

- conditions in global financial markets and domestic and international economic and social conditions, including inflation, risk of recession, political uncertainty and discord, geopolitical events or conflicts, international trade policies and sanctions laws;
- the impact of the introduction of or any changes in laws, regulations, rules or government policies with respect to financial markets, climate change, increased regulatory scrutiny or enforcement actions and our ability to comply with these requirements;
- volatility in commodity prices and equity prices, and price volatility of financial benchmarks and instruments such as interest rates, credit spreads, equity indices, foreign exchange rates, and mortgage origination trends;
- the impact of climate change and the transition to renewable energy;
- the business environment in which we operate and trends in our industries, including trading volumes, prevalence of clearing, demand for data services, mortgage lending activity, fees, changing regulations, competition and consolidation;
- our ability to minimize the risks associated with operating clearing houses in multiple jurisdictions;
- our exchanges’ and clearing houses’ compliance with their respective regulatory and oversight responsibilities;
- the resilience of our electronic platforms and soundness of our business continuity and disaster recovery plans;
- our ability to realize the expected benefits of our acquisitions and our investments, including our ability to close the Black Knight acquisition on the terms and timing expected;
- our ability to execute our growth strategy, identify and effectively pursue, implement and integrate acquisitions and strategic alliances and realize the synergies and benefits of such transactions within the expected time frame;
- the performance and reliability of our trading, clearing and mortgage technologies and those of third-party service providers;

- our ability to keep pace with technological developments and client preferences;
- our ability to ensure that the technology we utilize is not vulnerable to cyberattacks, hacking and other cybersecurity risks or other disruptive events or to minimize the impact of any such events;
- our ability to keep information and data relating to the customers of the users of the software and services provided by our ICE Mortgage Technology business confidential;
- the impacts of a public health emergency or pandemic, including the re-emergence of the COVID-19 pandemic, on our business, results of operations and financial condition as well as the broader business environment;
- our ability to identify trends and adjust our business to benefit from such trends, including trends in the U.S. mortgage industry such as inflation rates, interest rates, new home purchases, refinancing activity, and home builder and buyer sentiment, among others;
- our ability to evolve our benchmarks and indices in a manner that maintains or enhances their reliability and relevance;
- the accuracy of our cost and other financial estimates and our belief that cash flows from operations will be sufficient to service our debt and to fund our operational and capital expenditure needs;
- our ability to incur additional debt and pay off our existing debt in a timely manner;
- our ability to maintain existing market participants and data and mortgage technology customers, and to attract new ones;
- our ability to offer additional products and services, leverage our risk management capabilities and enhance our technology in a timely and cost-effective fashion;
- our ability to attract, develop and retain key talent;
- our ability to protect our intellectual property rights and to operate our business without violating the intellectual property rights of others; and
- potential adverse results of threatened or pending litigation and regulatory actions and proceedings.

These risks and other factors include, among others, those set forth in Part 1, Item 1(A) under the caption “Risk Factors” in our 2022 Form 10-K, as filed with the SEC on February 2, 2023. Due to the uncertain nature of these factors, management cannot assess the impact of each factor on the business or the extent to which any factor, or combination of factors, may cause actual results to differ materially from those contained in any forward-looking statements.

Any forward-looking statement speaks only as of the date on which such statement is made, and we undertake no obligation to update any of these statements to reflect events or circumstances occurring after the date of this Quarterly Report. New factors may emerge and it is not possible to predict all factors that may affect our business and prospects.

Overview

We are a provider of market infrastructure, data services and technology solutions to a broad range of customers including financial institutions, corporations and government entities. Our products, which span major asset classes including futures, equities, fixed income and residential mortgages in the U.S., provide our customers with access to mission critical tools that are designed to increase asset class transparency and workflow efficiency. Although we report our results in three reportable business segments, we operate as one business, leveraging the collective expertise, particularly in data services and technology, that exists across our platforms to inform and enhance our operations. Our segments are as follows:

- **Exchanges:** We operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities.
- **Fixed Income and Data Services:** We provide fixed income pricing, reference data, indices, analytics and execution services as well as global CDS clearing and multi-asset class data delivery solutions.
- **Mortgage Technology:** We provide a technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

Recent Developments

Pending Acquisition of Black Knight, Inc.

On May 4, 2022, we announced that we had entered into a definitive agreement to acquire Black Knight, Inc., or Black Knight, a software, data and analytics company that serves the housing finance continuum, including real estate data,

mortgage lending and servicing, as well as the secondary markets. Pursuant to the merger agreement, Sub will merge with and into Black Knight, with Black Knight surviving as a wholly owned subsidiary of ICE.

On March 7, 2023, ICE and Black Knight announced that, in connection with the merger agreement, Black Knight has entered into an agreement to sell its Empower loan origination business. On March 7, 2023, ICE and Black Knight also entered into an amendment to the merger agreement to reduce the per share merger consideration to be paid by ICE at the effective time of the merger. As of March 7, 2023, the amended transaction was valued at approximately \$11.7 billion, or \$75 per share of Black Knight common stock, with cash comprising 90% of the value of the aggregate transaction consideration and shares of our common stock comprising 10% of the value of the aggregate transaction consideration. The aggregate cash component of the transaction consideration is fixed at \$10.5 billion, and the value of the aggregate stock component of the transaction consideration will fluctuate with the market price of our common stock and will be determined based on the average of the volume weighted averages of the trading prices of our common stock on each of the ten consecutive trading days ending three trading days prior to the closing of the merger. If consummated, we expect that this transaction will build on our position as a provider of end-to-end electronic workflow solutions for the rapidly evolving U.S. residential mortgage industry. We believe the Black Knight ecosystem adds value for clients of all sizes across the mortgage and real estate lifecycles by helping organizations lower costs, increase efficiencies, grow their businesses, and reduce risk.

On March 9, 2023, the FTC filed an administrative complaint alleging that the proposed transaction between ICE and Black Knight, if consummated, would be an unfair method of competition in violation of Section 5 of the Federal Trade Commission Act, and that it would substantially lessen competition, or tend to create a monopoly, in violation of Section 7 of the Clayton Act. The complaint seeks a variety of injunctive relief, including, among other things, a prohibition on the completion of the transaction without the FTC's consent and, if the transaction is completed, a divestiture or reconstitution of assets in a manner that restores such separate and independent businesses as the parties had operated prior to the completion of the transaction. On April 10, 2023, the FTC filed a complaint in the United States District Court for the Northern District of California for a temporary restraining order and preliminary injunction enjoining the completion of the transaction. On April 21, 2023, the court entered a temporary restraining order enjoining the completion of the transaction until the court rules on the FTC's motion for a preliminary injunction. In their answers to the administrative and court complaints, filed on March 20, 2023 and April 25, 2023, respectively, ICE and Black Knight denied the FTC's substantive allegations; asserted numerous affirmative defenses; described the pro-competitive aspects and significant lender, servicer, investor, vendor and consumer benefits relating to this transaction; and denied that the combination of their respective businesses would violate any laws. Additionally, the answers to the court complaint contained counterclaims by ICE and Black Knight against the FTC seeking declaratory relief that the FTC's administrative process is unconstitutional and should be enjoined. We are vigorously defending against the FTC's administrative and court complaints.

On March 30, 2023, our amended proxy statement/prospectus on Form S-4 was declared effective by the SEC, and on April 28, 2023, Black Knight stockholders approved the amendment to the merger agreement.

On July 17, 2023, we announced that we have agreed to sell Black Knight's Optimal Blue business to the same purchaser as that of the Empower business for \$700 million. The structure of the proposed transaction includes a payment of \$200 million in cash, with the remainder financed by a \$500 million promissory note issued by the purchaser to Black Knight, as a subsidiary of ICE, at the closing of the transaction. The closing of the proposed Optimal Blue divestiture transaction is subject to the closing of our acquisition of Black Knight, the closing of the acquisition of Black Knight's Empower loan origination system business, and the satisfaction of other customary closing conditions.

The transaction is expected to close in the second half of 2023 following the receipt of regulatory approvals, a favorable resolution of the FTC litigation concerning this transaction, and the satisfaction of customary closing conditions. See Note 13 in our consolidated financial statements in this Quarterly Report where additional details of this transaction are discussed.

Global Market Conditions

Our results of operations are affected by global economic conditions, including macroeconomic conditions and geopolitical events or conflicts. During 2022 and into the first half of 2023, macroeconomic conditions, including rising interest rates, inflation and significant market volatility, along with geopolitical concerns, including the conflict in Ukraine and the sanctions and other measures that have been and continue to be imposed in response to the conflict, have created ongoing uncertainty and volatility in the global economy and resulted in a dynamic operating environment.

Our business has been impacted positively and negatively by these global economic conditions. For instance, due to market volatility and rising interest rates, we have seen increased trading across a number of our products, such as interest rate and equity futures, credit default swaps and bonds. Conversely, increases in mortgage interest rates in 2022 and the first half of 2023 have resulted in reduced consumer and investor demand for mortgages and adversely impacted

the transaction-based revenues in our Mortgage Technology segment. If mortgage rates remain high or further increase, or if banks change their mortgage lending practices, our Mortgage Technology segment revenues may be further impacted.

We have continued suspension of all services in Russia except for limited offerings to non-sanctioned entities. From an operational perspective, our businesses, including our exchanges, clearing houses, listings venues, data services businesses and mortgage platforms, have not suffered a material negative impact as a result of the events in Ukraine and the surrounding region.

We expect the macroeconomic environment to remain dynamic in the near-term, and we continue to monitor macroeconomic conditions, including interest rates, the inflationary environment, geopolitical events and military conflicts, including repercussions from the conflict in Ukraine, and the impact that any of the foregoing may have on the global economy and on our business. During the first half of 2023, we closely monitored the credit worthiness of our counterparties and investment agents during the recent banking sector events, scrutinized counterparties directly impacted and monitored for any potential contagion. We did not suffer any material negative impact from the banking sector events that occurred during the first half of 2023. In light of the current and expected macroeconomic environment we will continue to closely monitor credit worthiness of our counterparties, clearing members and our financial service providers and take risk management measures in line with established risk management frameworks.

Regulation

Our activities and the markets in which we operate are subject to regulations that impact us as well as our customers, and, in turn, meaningfully influence our activities, the manner in which we operate and our strategy. We are primarily subject to the jurisdiction of regulatory agencies in the U.S., U.K., EU, Canada, Singapore and Abu Dhabi. Failure to satisfy regulatory requirements can or may give rise to sanctions by the applicable regulator.

Global policy makers have undertaken reviews of their existing legal framework governing financial markets in connection with regulatory reform, and have either passed new laws and regulations, or are in the process of debating and/or enacting new laws and regulations that apply to our business and to our customers' businesses. Legislative and regulatory actions may impact the way in which we or our customers conduct business and may create uncertainty, which could affect trading volumes or demand for market data. See Part 1, Item 1 "Business — Regulation" and Part 1, Item 1(A) "Risk Factors" included in our 2022 Form 10-K for a discussion of the primary regulations applicable to our business and certain risks associated with those regulations.

Domestic and foreign policy makers continue to review their legal frameworks governing financial markets, and periodically change the laws and regulations that apply to our business and to our customers' businesses. Our key areas of focus on these evolving efforts are:

- **Policy intervention to address high energy prices.** Various legislative proposals in the European Union, or EU, have been adopted to address high energy prices and impact ICE Endex, the primary European exchange for the benchmark European gas contract, and ICE Clear Europe, which clears ICE Endex contracts. These policy interventions include price limiting mechanisms for exchange-traded gas products and a new liquified natural gas, or LNG, import benchmark. In December 2022, the EU adopted a price cap on certain Dutch Title Transfer Facility, or TTF, derivatives traded on ICE Endex effective February 2023. In March 2023, the European Commission extended the price cap to derivatives on all other EU gas hubs effective May 2023. In December 2022, a coalition of G7 and other nations set the price of certain Russian crude oil at or below \$60 a barrel, which impacts the services we offer to clients. Global leaders continue to discuss the implementation of additional sanctions against Russia.
- **Changes to EU regulation of gas and power markets.** In March 2023, the European Commission published legislative proposals to amend the Regulation on Wholesale Energy Market Integrity and Transparency, or REMIT, by introducing requirements for non-EU firms trading in European gas and power markets establishing an office in the EU. These requirements could make trading on ICE Endex more difficult and could result in a reduction in volumes and liquidity. We are monitoring the impact of these proposals on ICE Endex.
- **EMIR 3.0.** In December 2022, the European Commission proposed amendments to the European Market Infrastructure Regulation, or EMIR, requiring certain EU market participants to clear specified derivative transactions at an EU central counterparty. The European Commission has identified three classes of derivatives as being of substantial systemic importance, including short-term interest rate derivatives, which are traded on ICE Futures Europe and cleared at ICE Clear Europe, and euro denominated credit default swaps cleared at ICE Clear Credit. If adopted, the proposal could result in a reduction of the cleared volume of these contracts at ICE Clear Europe and ICE Clear Credit. We are monitoring the impact of this proposal on ICE Clear Europe and ICE Clear Credit.

- **Benchmarks Regulation.** ICE Benchmark Administration, Limited, or IBA, the administrator of London Interbank Offered Rate, or LIBOR, is no longer publishing any LIBOR settings using panel bank contributions after the end of June 2023. The Financial Conduct Authority, or FCA, has decided to use its legal powers under the U.K. Benchmarks Regulation, or U.K. BMR, to require IBA to continue publishing 3-month "synthetic" Sterling LIBOR until the end of March 2024 and 1-, 3- and 6- month "synthetic" U.S. Dollar LIBOR until the end of September 2024. "Synthetic" LIBOR settings are not based on panel bank contributions and are not representative of the underlying market or economic reality the settings were previously intended to measure. All other LIBOR settings have ceased to be published. Usage of "synthetic" LIBOR settings may be restricted or prohibited in certain circumstances under applicable law.

Finally, the European Commission used its powers under the EU Benchmarks Regulation, or EU BMR, to designate replacement benchmarks for certain Swiss franc LIBOR settings. Certain benchmarks provided by our index provider businesses may continue to be used by supervised entities in the EU under EU BMR transitional provisions. Although these are currently scheduled to end December 31, 2023, the European Commission adopted a delegated act on July 14, 2023, to extend the transition period until December 31, 2025. Also, the European Commission has published a consultation on the scope of the EU BMR and the regime for third-country benchmark administrators in preparation of the development of a legislative proposal which is not expected until late 2023. Uncertainties relating to the extension of the transition period and the scope of the EU BMR may impact our ability to provide benchmarks into the EU.
- **EU Deforestation Regulation.** The EU Deforestation Regulation, or EUDR, aims to curb the EU market's impact on global deforestation and forest degradation by requiring seven commodities, including cocoa and coffee, and certain specified products made from them, to be deforestation-free to be sold on the EU market or exported from it. The EUDR obligations are scheduled to be effective in December 2024. The EUDR requirements may decelerate the physical trade of cocoa and coffee, impact the useable EU coffee and cocoa physical inventories, and reduce trading volumes on ICE Futures Europe of the Robusta Coffee Contract and London Cocoa Contract and on ICE Futures US of the Coffee C ® Contract (Arabica). We are monitoring the impact of the EUDR and working to implement the regulation.

Consolidated Financial Highlights

The following summarizes our results and significant changes in our consolidated financial performance for the periods presented (dollars in millions, except per share amounts).

 21724	 21726	 21727
 21730	 21731	 21732

⁽¹⁾ Operating loss from our Mortgage Technology segment was \$45 million for the six months ended June 30, 2023.
⁽²⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. Adjusted net income attributable to ICE is presented net of taxes. These adjusted numbers are not calculated in accordance with U.S. GAAP. See “—Non-GAAP Financial Measures” below.

	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Revenues, less transaction-based expenses	\$ 3,784	\$ 3,713	2 %	\$ 1,888	\$ 1,814	4 %
Recurring revenues ⁽¹⁾	\$ 1,908	\$ 1,851	3 %	\$ 955	\$ 930	2 %
Transaction revenues, net ⁽¹⁾	\$ 1,876	\$ 1,862	1 %	\$ 933	\$ 884	6 %
Operating expenses	\$ 1,860	\$ 1,852	— %	\$ 933	\$ 945	(1) %
Adjusted operating expenses ⁽²⁾	\$ 1,496	\$ 1,486	1 %	\$ 756	\$ 740	2 %
Operating income	\$ 1,924	\$ 1,861	3 %	\$ 955	\$ 869	10 %
Adjusted operating income ⁽²⁾	\$ 2,288	\$ 2,227	3%	\$ 1,132	\$ 1,074	5%
Operating margin	51 %	50 %	1 pt	51 %	48 %	3 pts
Adjusted operating margin ⁽²⁾	60 %	60 %	—	60 %	59 %	1 pt
Other income/(expense), net	\$ (228)	\$ (290)	(21) %	\$ (108)	\$ (130)	(16) %
Income tax expense	\$ 207	\$ 338	(39) %	\$ 32	\$ 173	(82) %
Effective tax rate	12 %	22 %	(10 pts)	4 %	23 %	(19 pts)
Net income attributable to ICE	\$ 1,454	\$ 1,212	20 %	\$ 799	\$ 555	44 %
Adjusted net income attributable to ICE ⁽²⁾	\$ 1,593	\$ 1,543	3 %	\$ 802	\$ 739	8 %
Diluted earnings per share attributable to ICE common stockholders	\$ 2.59	\$ 2.16	20 %	\$ 1.42	\$ 0.99	43 %
Adjusted diluted earnings per share attributable to ICE common stockholders ⁽²⁾	\$ 2.84	\$ 2.75	3 %	\$ 1.43	\$ 1.32	8 %
Cash flows from operating activities	\$ 1,805	\$ 1,725	5 %			
Free cash flow ⁽³⁾	\$ 1,602	\$ 1,521	5 %			
Adjusted free cash flow ⁽³⁾	\$ 1,652	\$ 1,406	18 %			

⁽¹⁾ We define recurring revenues as the portion of our revenues that are generally predictable, stable, and can be expected to occur at regular intervals in the future with a relatively high degree of certainty and visibility. We define transaction revenues as those associated with a more specific point-in-time service, such as a trade execution.

⁽²⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. Adjusted net income attributable to ICE and adjusted diluted earnings per share attributable to ICE common stockholders are presented net of taxes. These adjusted figures are not calculated in accordance with U.S. GAAP. See “—Non-GAAP Financial Measures” below.

⁽³⁾ We believe these non-GAAP liquidity measures provide useful information to management and investors to analyze cash resources generated from our operations. We believe that free cash flow is useful as one of the bases for comparing our performance with our competitors, and demonstrates our ability to convert the reinvestment of capital expenditures and capitalized software development costs required to maintain and grow our business, and that adjusted free cash flow eliminates the impact of timing differences related to the payment of section 31 fees. These figures are not calculated in accordance with U.S. GAAP. See “—Non-GAAP Liquidity Measures” below.

- Revenues, less transaction-based expenses, increased \$71 million and \$74 million for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. See “—Exchanges Segment”, “—Fixed Income and Data Services Segment” and “—Mortgage Technology Segment” below for a discussion of the significant changes in our revenues. The increase in revenues during the six months ended June 30, 2023 includes \$19 million in unfavorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable period in 2022 and the increase in revenues during the three months ended June 30, 2023 includes \$3 million in favorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable period in 2022. See Item 3 “Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk” below for additional information on the impact of currency fluctuations.
- Operating expenses increased \$8 million for the six months ended June 30, 2023, and decreased \$12 million for the three months ended June 30, 2023, from the comparable periods in 2022. See “—Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses. The increase in operating expenses during the six months ended June 30, 2023 includes \$8 million in favorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable period in 2022 and the decrease in operating expenses during the three months ended June 30, 2023 do not include any foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable period in 2022. See Item 3 “Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk” below for additional information on the impact of currency fluctuations.

Variability in Quarterly Comparisons

Our business environment has been characterized by:

- globalization of marketplaces, customers and competitors;
- growing customer demand for workflow efficiency and automation;
- commodity, interest rate, inflation rate and financial markets volatility and uncertainty;
- growing demand for data to inform customers' risk management and investment decisions;
- evolving, increasing and disparate regulation across multiple jurisdictions;
- price volatility increasing customers' demand for risk management services;
- increasing focus on capital and cost efficiencies;
- customers' preference to manage risk in markets demonstrating the greatest depth of liquidity and product diversity;
- the evolution of existing products and new product innovation to serve emerging customer needs and changing industry agreements;
- rising demand for speed, data, data capacity and connectivity by market participants, necessitating increased investment in technology; and
- consolidation and increasing competition among global markets for trading, clearing and listings.

For additional information regarding the factors that affect our results of operations, see Item 1(A) "Risk Factors" included in our 2022 Form 10-K.

Segment Results

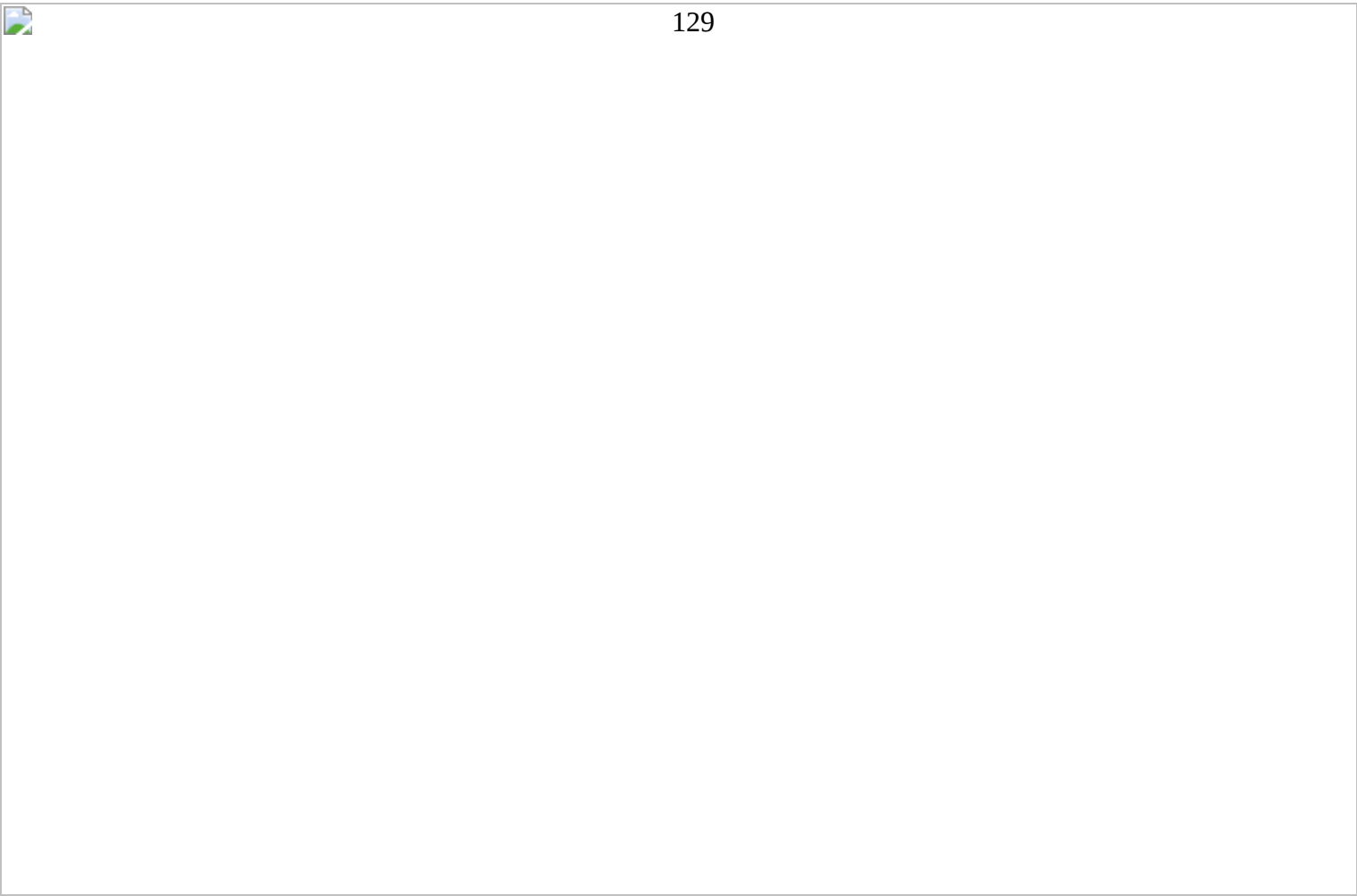
Our business is conducted through three reportable business segments:

- **Exchanges:** We operate regulated marketplaces for the listing, trading and clearing of a broad array of derivatives contracts and financial securities;
- **Fixed Income and Data Services:** We provide fixed income pricing, reference data, indices, analytics and execution services as well as global CDS clearing and multi-asset class data delivery solutions; and
- **Mortgage Technology:** We provide a technology platform that offers customers comprehensive, digital workflow tools that aim to address the inefficiencies that exist in the U.S. residential mortgage market, from application through closing and the secondary market.

While revenues are recorded specifically in the segment in which they are earned or to which they relate, a significant portion of our operating expenses are not solely related to a specific segment because the expenses serve functions that are necessary for the operation of more than one segment. We directly allocate expenses when reasonably possible to do so. Otherwise, we use a pro-rata revenue approach as the allocation method for the expenses that do not relate solely to one segment and serve functions that are necessary for the operation of all segments. Our segments do not engage in intersegment transactions.

Exchanges Segment

The following presents selected statements of income data for our Exchanges segment (dollars in millions and YTD represents the six-month periods ended June 30th):



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⁽¹⁾ The adjusted figures in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted figures are not calculated in accordance with U.S. GAAP. See “ —Non-GAAP Financial Measures” below.

	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change*	2023	2022	Change
Revenues:						
Energy futures and options	\$ 700	\$ 618	13 %	\$ 355	\$ 265	34 %
Agricultural and metals futures and options	147	122	20	77	61	27
Financial futures and options	232	253	(8)	104	123	(16)
Futures and options	1,079	993	9	536	449	19
Cash equities and equity options	1,215	1,357	(10)	544	698	(22)
OTC and other	205	205	—	104	108	(3)
Transaction and clearing, net	2,499	2,555	(2)	1,184	1,255	(6)
Data and connectivity services	463	432	7	231	218	6
Listings	252	260	(3)	126	131	(5)
Revenues	3,214	3,247	(1)	1,541	1,604	(4)
Transaction-based expenses ⁽¹⁾	1,024	1,159	(12)	448	599	(25)
Revenues, less transaction-based expenses	2,190	2,088	5	1,093	1,005	9
Other operating expenses	508	484	5	249	244	3
Depreciation and amortization	123	118	4	62	60	3
Acquisition-related transaction and integration costs	—	1	n/a	—	—	n/a
Operating expenses	631	603	5	311	304	2
Operating income	\$ 1,559	\$ 1,485	5 %	\$ 782	\$ 701	12 %
Recurring revenues	\$ 715	\$ 692	3 %	\$ 357	\$ 349	2 %
Transaction revenues, net	\$ 1,475	\$ 1,396	6 %	\$ 736	\$ 656	12 %

⁽¹⁾Transaction-based expenses are largely attributable to our cash equities and options business.

*Percentage changes in the table above deemed "n/a" are not meaningful.

Exchanges Revenues

Our Exchanges segment includes transaction and clearing revenues from our futures and NYSE exchanges, related data and connectivity services, and our listings business. Transaction and clearing revenues consist of fees collected from derivatives, cash equities and equity options trading and derivatives clearing, and are reported on a net basis, except for the NYSE transaction-based expenses discussed below. Rates per-contract, or RPC, are driven by the number of contracts or securities traded and the fees charged per contract, net of certain rebates. Our per-contract transaction and clearing revenues will depend upon many factors, including, but not limited to, market conditions, transaction and clearing volume, product mix, pricing, applicable revenue sharing and market making agreements, and new product introductions.

Transaction and clearing revenues are generally assessed on a per-contract basis and revenues and profitability fluctuate with changes in contract volume and product mix. We consider data and connectivity services revenues and listings revenues to be recurring revenues. Our data and connectivity services revenues are recurring subscription fees related to the various data and connectivity services that we provide which are directly attributable to our exchange venues. Our listings revenues are also recurring subscription fees that we earn for the provision of NYSE listings services for public companies and ETFs, and related corporate actions for listed companies.

For both the six months ended June 30, 2023 and 2022, 19% of our Exchanges segment revenues, less transaction-based expenses, were billed in pounds sterling or euros and for the three months ended June 30, 2023 and 2022, 19% and 18%, respectively, of our Exchanges segment revenues, less transaction-based expenses, were billed in pounds sterling or euros. Due to the fluctuations of the pound sterling and euro compared to the U.S. dollar, our Exchanges segment revenues, less transaction-based expenses, were lower by \$15 million for the six months ended June 30, 2023, and higher by \$2 million for the three months ended June 30, 2023, from the comparable periods in 2022.

Our exchange transaction and clearing revenues are presented net of rebates. We recorded rebates of \$488 million and \$464 million for the six months ended June 30, 2023 and 2022, respectively, and \$236 million and \$201 million for the three months ended June 30, 2023 and 2022, respectively. We offer rebates in certain of our markets primarily to support market liquidity and trading volume by providing qualified participants in those markets a discount to the applicable commission rate. Such rebates are calculated based on volumes traded. The increase in rebates for the six

months and three months ended June 30, 2023 is primarily due to higher volumes traded as compared to the comparable periods in 2022.

- **Energy Futures and Options:** Total energy volume increased 5% and revenues increased 13% for the six months ended June 30, 2023 from the comparable period in 2022 and volume increased 16% and revenues increased 34% for the three months ended June 30, 2023 from the comparable period in 2022.
 - Total oil futures and options volume increased 5% and 26% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, primarily due to reduced uncertainty, normalization of price levels, and increased focus on Brent with Midland WTI now deliverable into the Brent Basket, providing additional physical liquidity and exposure.
 - Our global natural gas futures and options volume increased 6% and 3% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, primarily due to strength across our Dutch TTF complex as the second quarter of 2023 benefited from reduced uncertainty and normalization of price levels.
 - Our environmentals and other futures and options volume decreased 5% and 6% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, due in part to lower power volumes in the second quarter of 2023 versus the year ago period.
- **Agricultural and Metals Futures and Options:** Total volumes in our agricultural and metals futures and options markets increased 23% and 27% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022 and revenues increased 20% and 27% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. The first half of 2023 benefited from price volatility as a result of weather-related supply and demand dynamics driving an increased need to manage risk across our commodity markets.
 - Sugar futures and options volumes increased 30% and 37% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022.
 - Other agricultural and metal futures and options volume increased 17% and 19% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022.
- **Financial Futures and Options:** Total volumes in our financial futures and options markets were flat for the six months ended June 30, 2023, and decreased 13% for the three months ended June 30, 2023, from the comparable periods in 2022 and revenues decreased 8% and 16% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, including the impacts of foreign exchange effects. The second quarter of 2022 benefited from elevated volatility across global markets driven by geopolitical events, central bank activity and inflationary concerns.
 - Interest rate futures and options volume increased 4% for the six months ended June 30, 2023, and decreased 12% for the three months ended June 30, 2023, respectively, from the comparable periods in 2022, and revenue decreased 6% and 15% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. The second quarter of 2022 benefited from elevated interest rate volatility related to increased speculation regarding central bank activity due to inflation concerns. Interest rate futures and options revenues were \$149 million and \$158 million for the six months ended June 30, 2023 and 2022, respectively, and \$65 million and \$76 million for the three months ended June 30, 2023 and 2022, respectively.
 - Other financial futures and options volume, which includes our MSCI®, FTSE® and NYSE FANG+ equity index products, decreased 13% and 21% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. Financial futures and options revenue decreased 12% and 17% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022 as the first half of 2022 benefited from heightened volatility related to the conflict in Ukraine. Other financial futures and options revenues were \$83 million and \$95 million for the six months ended June 30, 2023 and 2022, respectively, and \$39 million and \$47 million for the three months ended June 30, 2023 and 2022, respectively.
- **Cash Equities and Equity Options:** Cash equities volume decreased 13% and 16% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, due to lower total market volumes

as the first half of 2022 benefited from elevated volatility related to inflationary, recessionary and geopolitical concerns. Cash equities revenues, net of transaction-based expenses, were \$134 million and \$147 million for the six months ended June 30, 2023 and 2022, respectively, \$67 million and \$74 million for the three months ended June 30, 2023 and 2022, respectively. Equity options volume increased 3% and 1% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. The overall increase in equity options volume for the six months ended June 30, 2023 was driven by increased participation. Equity options revenues, net of transaction-based expenses, were \$57 million and \$51 million for the six months ended June 30, 2023 and 2022, respectively, and \$29 million and \$25 million for the three months ended June 30, 2023 and 2022, respectively.

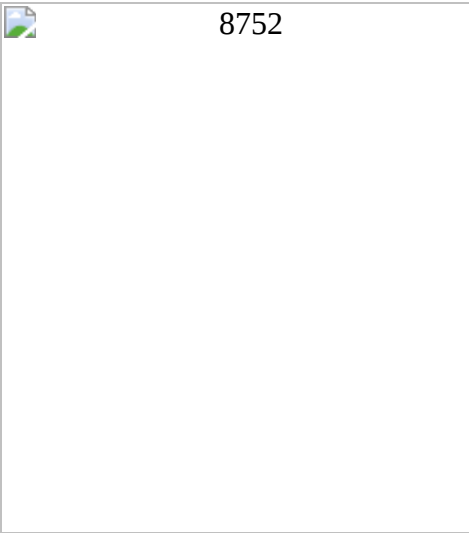
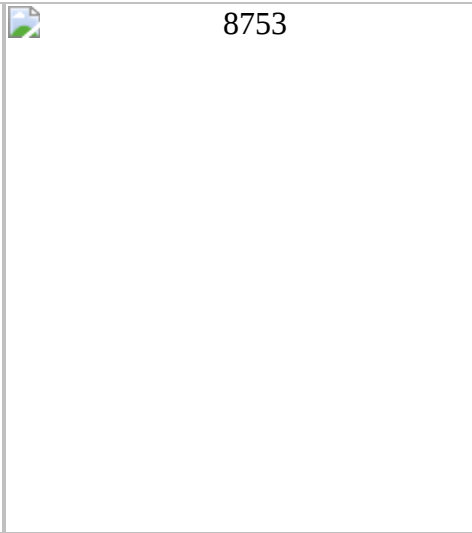
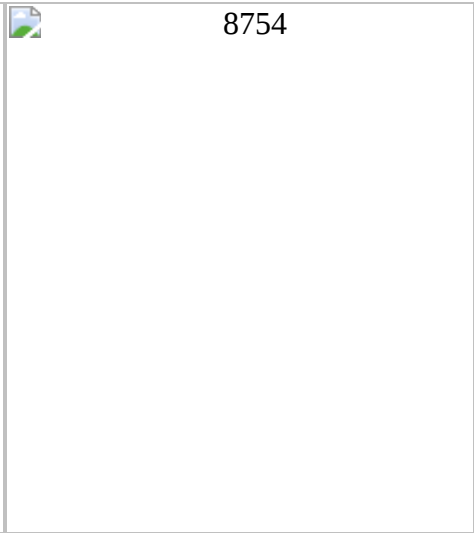
- **OTC and Other:** OTC and other transactions include revenues from our OTC energy business and other trade confirmation services, as well as interest income on certain clearing margin deposits, regulatory penalties and fines, fees for use of our facilities, regulatory fees charged to member organizations of our U.S. securities exchanges, designated market maker service fees, exchange membership fees and agricultural grading and certification fees. Our OTC and other revenues were flat for the six months ended June 30, 2023 and decreased 3% for the three months ended June 30, 2023, from the comparable periods in 2022, primarily due to a decrease in interest income on clearing margin deposits in the second quarter of 2023.
- **Data and Connectivity Services:** Our data and connectivity services revenues increased 7% and 6% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. The increase in revenue was driven by the strong retention rate of existing customers, the addition of new customers and increased purchases by existing customers.
- **Listings Revenues:** Through NYSE, NYSE American and NYSE Arca, we generate listings revenue related to the provision of listings services for public companies and ETFs, and related corporate actions for listed companies. Listings revenues decreased 3% and 5% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, driven by market volatility causing IPO delays. All listings fees are billed upfront and revenues are recognized over time as the identified performance obligations are satisfied.

Selected Operating Data

Volume of contracts traded, futures and options rate per contract and open interest are measures that we use in analyzing the performance of our futures and options contracts. Handled volume, matched volume and cash equities and equity options rate per contract are measures that we use in analyzing our NYSE cash equities and equity options performance. We believe each of these measures provides useful information for management and investors in understanding our performance. Management considers these metrics when making financial and operating decisions. Our calculation of these metrics may not be comparable to similarly titled measures used by other companies.

The following charts and tables present trading activity in our futures and options markets by commodity type based on the total number of contracts traded, as well as futures and options rate per contract (in millions, except for percentages and rate per contract amounts):

Volume and Rate per Contract

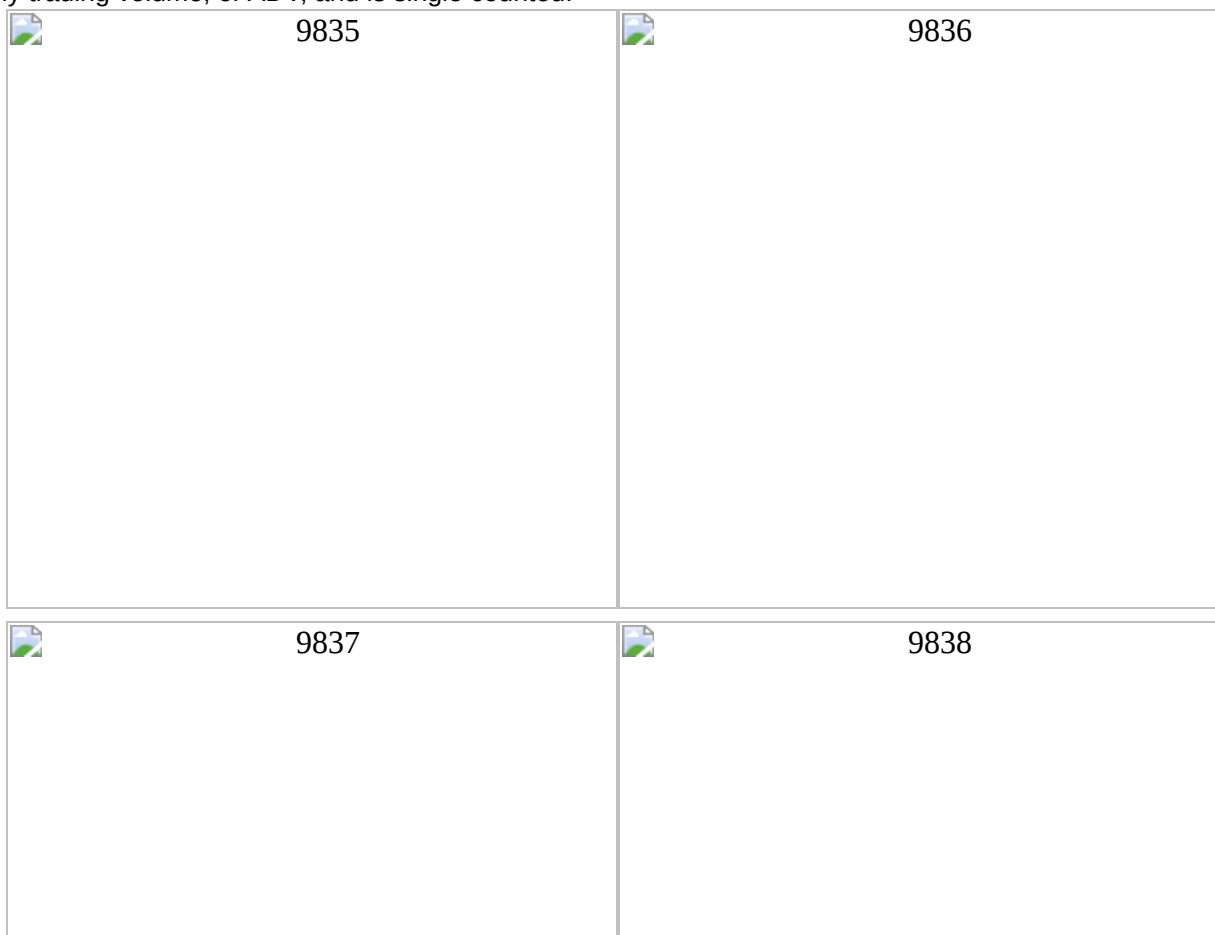
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	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Number of contracts traded (in millions):						
Energy futures and options	419	399	5 %	206	177	16 %
Agricultural and metals futures and options	63	52	23	33	26	27
Financial futures and options	330	328	—	137	158	(13)
Total	812	779	4 %	376	361	4 %
	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Average daily volume of contracts traded (in thousands):						
Energy futures and options	3,377	3,221	5 %	3,315	2,862	16 %
Agricultural and metals futures and options	510	416	23	530	416	27
Financial futures and options	2,605	2,599	—	2,182	2,523	(14)
Total	6,492	6,236	4 %	6,027	5,801	4 %
	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Rate per contract:						
Energy futures and options	\$ 1.67	\$ 1.55	8 %	\$ 1.73	\$ 1.49	16 %
Agricultural and metals futures and options	\$ 2.33	\$ 2.37	(2)%	\$ 2.36	\$ 2.36	— %
Financial futures and options	\$ 0.69	\$ 0.76	(9)%	\$ 0.75	\$ 0.77	(3)%

Open Interest

	As of June 30,		
	2023	2022	Change
Open interest — in thousands of contracts:			
Energy futures and options	49,247	43,985	12 %
Agricultural and metals futures and options	4,182	3,519	19
Financial futures and options	20,995	28,129	(25)
Total	74,424	75,633	(2)%

The following charts and tables present selected cash and equity options trading data. All trading volume below is presented as average net daily trading volume, or ADV, and is single counted:



	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
NYSE cash equities (shares in millions):						
Total cash handled volume	2,275	2,606	(13)%	2,169	2,591	(16)%
Total cash market share matched	19.8 %	20.1 %	(0.3 pts)	19.8 %	20.2 %	(0.4 pts)
NYSE equity options (contracts in thousands):						
NYSE equity options volume	8,204	7,937	3 %	7,701	7,647	1 %
Total equity options volume	40,839	38,350	6 %	39,244	36,672	7 %
NYSE share of total equity options	20.1 %	20.7 %	(0.6 pts)	19.6 %	20.9 %	(1.3 pts)
Revenue capture or rate per contract:						
Cash equities rate per contract (per 100 shares)	\$0.048	\$0.046	4 %	\$0.050	\$0.047	7 %
Equity options rate per contract	\$0.06	\$0.05	10 %	\$0.06	\$0.05	20 %

Handled volume represents the total number of shares of equity securities, ETFs and crossing session activity internally matched on our exchanges or routed to and executed on an external market center. Matched volume represents the total number of shares of equity securities, ETFs and crossing session activity executed on our exchanges.

Transaction-Based Expenses

Our equities and equity options markets pay fees to the SEC pursuant to Section 31 of the Exchange Act. Section 31 fees are recorded on a gross basis as a component of transaction and clearing fee revenue. These Section 31 fees are assessed to recover the government’s costs of supervising and regulating the securities markets and professionals and are subject to change. We, in turn, collect corresponding activity assessment fees from member organizations clearing or settling trades on the equities and options exchanges, and recognize these amounts in our transaction and clearing revenues when invoiced. The activity assessment fees are designed to equal the Section 31 fees. As a result, activity assessment fees and the corresponding Section 31 fees do not have an impact on our net income, although the timing of payment by us will vary from collections. Section 31 fees were \$175 million and \$174 million for the six months ended June 30, 2023 and 2022, respectively, and \$56 million and \$123 million for the three months ended June 30, 2023 and 2022, respectively. The decrease in Section 31 fees during the three months ended June 30, 2023 was primarily due to a decrease in rates. The fees we collect are included in cash at the time of receipt and we remit the amounts to the SEC semi-annually as required. The total amount is included in current liabilities and was \$173 million as of June 30, 2023.

We make liquidity payments to cash and options trading customers, as well as routing charges made to other exchanges which are included in transaction-based expenses. We incur routing charges when we do not have the best bid or offer in the market for a security that a customer is trying to buy or sell on one of our securities exchanges. In that case, we route the customer’s order to the external market center that displays the best bid or offer. The external market center charges us a fee per share (denominated in tenths of a cent per share) for routing to its system. We record routing charges on a gross basis as a component of transaction and clearing fee revenue. Cash liquidity payments, routing and clearing fees were \$849 million and \$985 million for the six months ended June 30, 2023 and 2022, respectively, and \$392 million and \$476 million for the three months ended June 30, 2023 and 2022, respectively.

Operating Expenses, Operating Income and Operating Margin

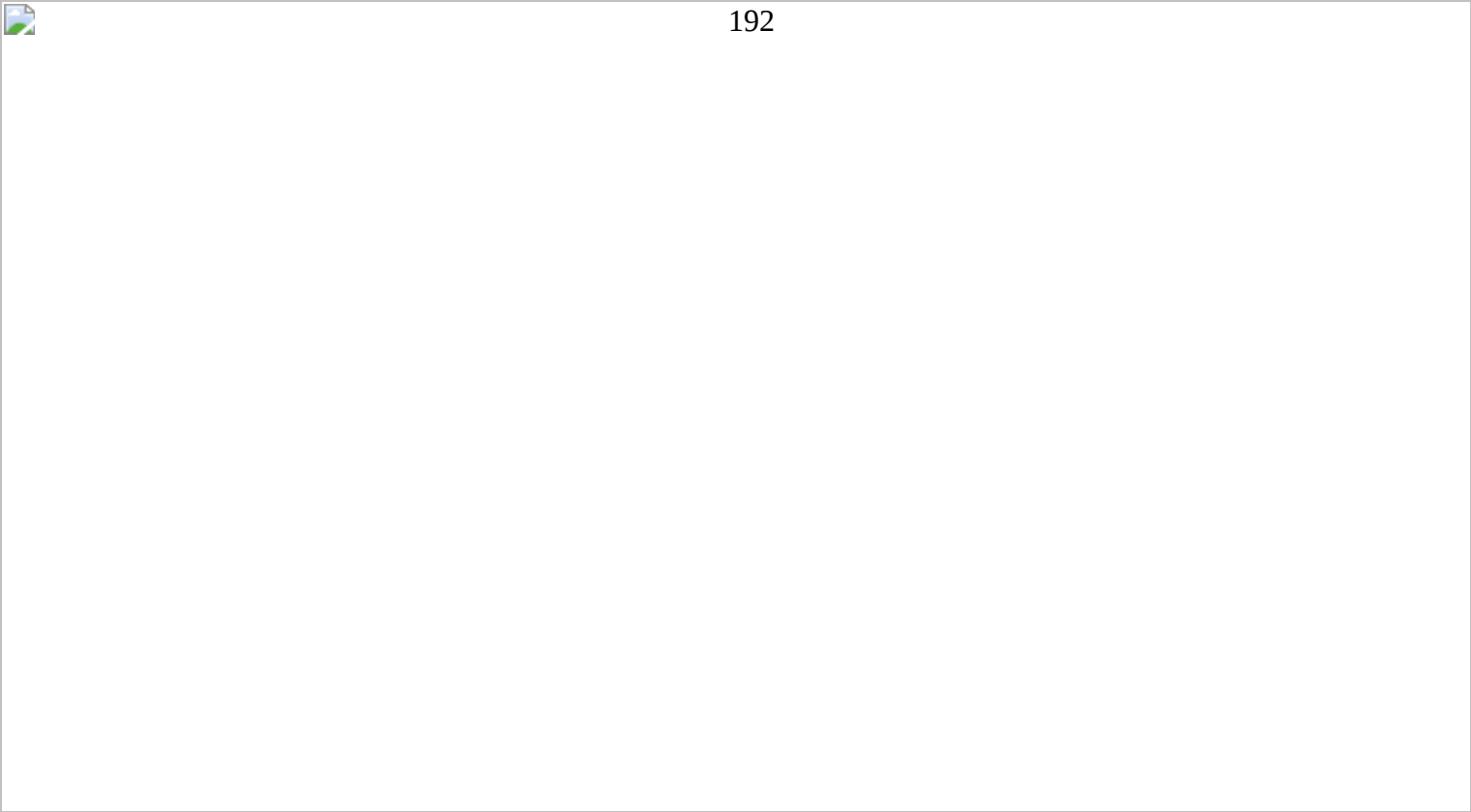
The following chart summarizes our Exchanges segment’s operating expenses, operating income and operating margin (dollars in millions). See “—Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses.

Exchanges Segment:	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Operating expenses	\$ 631	\$ 603	5 %	\$ 311	\$ 304	2 %
Adjusted operating expenses ⁽¹⁾	\$ 581	\$ 570	2 %	\$ 293	\$ 287	2 %
Operating income	\$ 1,559	\$ 1,485	5 %	\$ 782	\$ 701	12 %
Adjusted operating income ⁽¹⁾	\$ 1,609	\$ 1,518	6 %	\$ 800	\$ 718	11 %
Operating margin	71 %	71 %	—	72 %	70 %	2 pts
Adjusted operating margin ⁽¹⁾	73 %	73 %	—	73 %	71 %	2 pts

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted numbers are not calculated in accordance with U.S. GAAP. See “—Non-GAAP Financial Measures” below.

Fixed Income and Data Services Segment

The following charts and table present our selected statements of income data for our Fixed Income and Data Services segment (dollars in millions):



⁽¹⁾ The adjusted figures in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See “—Non-GAAP Financial Measures” below.

	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Revenues:						
Fixed income execution	\$ 60	\$ 40	51 %	\$ 28	\$ 25	17 %
CDS clearing	185	138	34	84	66	26
Fixed income data and analytics	553	551	—	277	274	1
Fixed income and credit	798	729	9	389	365	6
Other data and network services	311	292	7	157	147	7
Revenues	1,109	1,021	9	546	512	7
Other operating expenses	525	516	2	267	252	6
Depreciation and amortization	174	176	1	89	86	4
Operating expenses	699	692	1	356	338	5
Operating income	\$ 410	\$ 329	25 %	\$ 190	\$ 174	9 %
Recurring revenues	\$ 864	\$ 843	2 %	\$ 434	\$ 421	3 %
Transaction revenues	\$ 245	\$ 178	38 %	\$ 112	\$ 91	23 %

In the table above, we consider fixed income data and analytics revenues and other data and network services revenues to be recurring revenues.

For both the six months ended June 30, 2023 and 2022, 12% of our Fixed Income and Data Services segment revenues were billed in pounds sterling or euros and for both the three months ended June 30, 2023 and 2022, 12% of our Fixed Income and Data Services segment revenues were billed in pounds sterling or euros. As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues denominated in foreign currencies changes accordingly. Due to the fluctuations of the pound sterling and euro compared to the U.S. dollar, our Fixed Income and Data Services revenues were lower by \$4 million for the six months ended June 30, 2023, and higher by \$1 million for the three months ended June 30, 2023, than in the comparable periods in 2022.

Fixed Income and Data Services Revenues

Our Fixed Income and Data Services revenues increased 9% and 7% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. The increase in revenue was primarily due to strength in our fixed income execution CDS clearing business and our other data and network services.

- **Fixed Income Execution:** Fixed income execution includes revenues from ICE Bonds. Execution fees are reported net of rebates, which were nominal for both the six months and three months ended June 30, 2023 and 2022. Our fixed income execution revenues increased 51% and 17% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, due to increased activity as a result of continued interest rate volatility.
- **CDS Clearing:** CDS clearing revenues increased 34% and 26% for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022. The notional value of CDS cleared was \$10.2 trillion and \$13.6 trillion for the six months ended June 30, 2023 and 2022, respectively, and \$3.4 trillion and \$5.9 trillion for the three months ended June 30, 2023 and 2022, respectively. The increase in revenues was primarily due to net interest income on collateral balances.
- **Fixed Income Data and Analytics:** Our fixed income data and analytics revenues were flat for the six months ended June 30, 2023, and increased 1% for the three months ended June 30, 2023, from the comparable periods in 2022 due to strength in our index business in the second quarter of 2023.
- **Other Data and Network Services:** Our other data and network services revenues increased 7% for both the six months and three months ended June 30, 2023, from the comparable periods in 2022. The increase in revenues was driven by growth in our ICE Global Network offering, coupled with strength in our desktop, feeds and derivatives analytics revenues.

Annual Subscription Value, or ASV, represents, at a point in time, the data services revenues, which includes Fixed Income Data and Analytics as well as other data and network services, subscribed for the succeeding 12 months. ASV does not include new sales, contract terminations or price changes that may occur during that 12-month period. However,

while it is an indicative forward-looking metric, it does not provide a precise growth forecast of the next 12 months of data services revenues.

As of June 30, 2023, ASV was \$1.724 billion, which increased 3.9% compared to the ASV as of June 30, 2022. ASV represents nearly 100% of total data services revenues for this segment. This does not adjust for year-over-year foreign exchange fluctuations.

Operating Expenses, Operating Income and Operating Margin

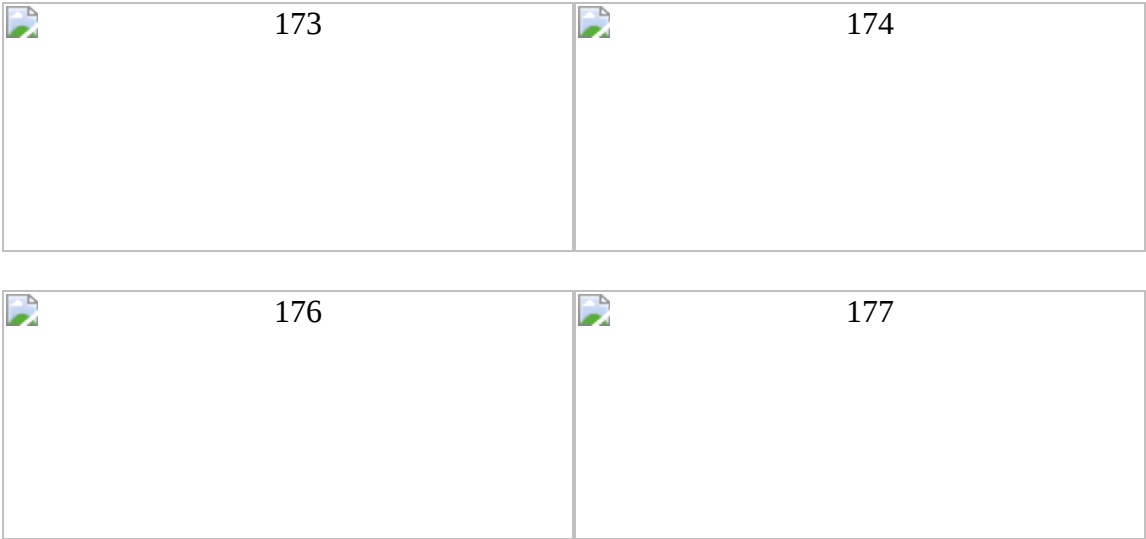
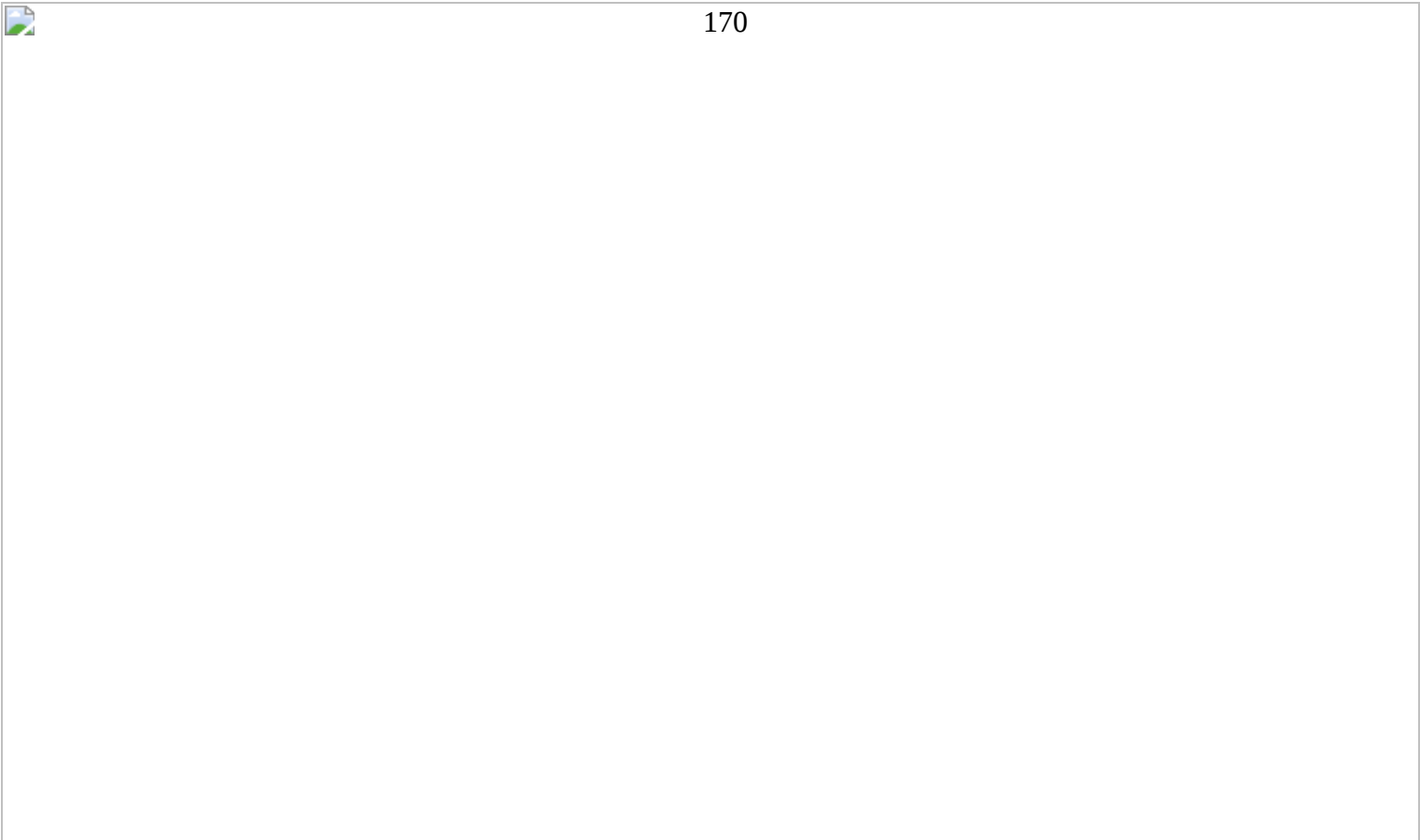
The following chart summarizes our Fixed Income and Data Services segment's operating expenses, operating income and operating margin (dollars in millions). See “—Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses.

Fixed Income and Data Services Segment:	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Operating expenses	\$ 699	\$ 692	1 %	\$ 356	\$ 338	5 %
Adjusted operating expenses ⁽¹⁾	\$ 614	\$ 599	3 %	\$ 313	\$ 294	7 %
Operating income	\$ 410	\$ 329	25 %	\$ 190	\$ 174	9 %
Adjusted operating income ⁽¹⁾	\$ 495	\$ 422	17 %	\$ 233	\$ 218	7 %
Operating margin	37 %	32 %	5 pts	35 %	34 %	1 pt
Adjusted operating margin ⁽¹⁾	45 %	41 %	4 pts	43 %	43 %	—

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted figures are not calculated in accordance with U.S. GAAP. See “—Non-GAAP Financial Measures” below.

Mortgage Technology Segment

The following charts and table present our selected statements of income data for our Mortgage Technology segment (dollars in millions):



⁽¹⁾ The adjusted figures in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted figures are not calculated in accordance with U.S. GAAP. See “—Non-GAAP Financial Measures” below.

	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Revenues:						
Origination technology	\$ 337	\$ 399	(15)%	\$ 170	\$ 196	(13)%
Closing solutions	87	138	(37)	47	66	(28)
Data and analytics	45	44	2	24	24	(5)
Other	16	23	(31)	8	11	(32)
Revenues	485	604	(20)	249	297	(16)
Other operating expenses	254	280	(9)	125	140	(11)
Depreciation and amortization	230	216	6	116	110	4
Acquisition-related transaction and integration costs	46	61	(24)	25	53	(53)
Operating expenses	530	557	(5)	266	303	(13)
Operating income/(loss)	\$ (45)	\$ 47	n/a	\$ (17)	\$ (6)	(158)%
Recurring revenues	\$ 329	\$ 316	4%	\$ 164	\$ 160	2%
Transaction revenues	\$ 156	\$ 288	(46)%	\$ 85	\$ 137	(38)%

*Percentage changes in the table above deemed "n/a" are not meaningful.

In the table above, we consider subscription fee and certain other revenues to be recurring revenues. Each revenue classification, above, contains a mix of recurring and transaction revenues, based on the various service offerings described in more detail, below.

Beginning in the first quarter of 2023, closing solutions revenues within our Mortgage Technology segment now include membership dues that were previously included in other revenues. We believe this is a more accurate reflection of the nature of these revenues. The impact of this change was not material, and the prior year periods have been adjusted for comparability.

Mortgage Technology Revenues

Our mortgage technology revenues are derived from our comprehensive, end-to-end U.S. residential mortgage platform. Our mortgage technology business is intended to enable greater workflow efficiency for customers focused on originating U.S. residential mortgage loans. Mortgage technology revenues decreased \$119 million and \$48 million for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022 primarily due to lower mortgage origination volumes driven by rising interest rates. See Note 6 of our consolidated financial statements in this Quarterly Report where discussed further.

- **Origination technology:** Our origination technology revenues decreased 15% and 13% during the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, due to lower transaction-based revenues as mortgage origination volumes declined. Our origination technology acts as a system of record for the mortgage transaction, automating the gathering, reviewing, and verifying of mortgage-related information and enabling automated enforcement of rules and business practices designed to help ensure that each completed loan transaction is of high quality and adheres to secondary market standards. These revenues are based on recurring Software as a Service, or SaaS, subscription fees, with an additive transaction-based or success-based pricing fee as lenders exceed the number of loans closed that are included with their monthly base subscription.

In addition, the ICE Mortgage Technology network provides originators connectivity to the mortgage supply chain and facilitates the secure exchange of information between our customers and a broad ecosystem of third-party service providers, as well as lenders and investors that are critical to consummating the millions of loan transactions that occur on our origination network each year. Revenue from the ICE Mortgage Technology network is largely transaction-based.

- **Closing solutions:** Our closing solutions revenues decreased 37% and 28% during the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, due to lower mortgage origination volumes. Our closing solutions connect key participants, such as lenders, title and settlement agents and individual county recorders, to digitize the closing and recording process. Closing solutions also include revenues from our MERSCORP Holdings, Inc., or MERS database, which provides a system of record for recording and tracking changes and servicing rights and beneficial ownership interests in loans secured by U.S. residential real estate. Revenues from closing solutions are largely transaction-based and are based on the volume of loans closed.

- Data and Analytics:** Our data and analytics revenues increased 2% for the six months ended June 30, 2023, and revenues decreased 5% during the three months ended June 30, 2023, from the comparable periods in 2022, due to lower transaction-based revenues. Revenues include those related to ICE Mortgage Technology's Data & Document Automation and Mortgage Analyzer solutions, or Analyzer (formerly known as AIQ), which offers customers greater efficiency by streamlining data collection and validation through our automated document recognition and data extraction capabilities. Analyzer revenues can be both recurring and transaction-based in nature. In addition, our data offerings include real-time industry and peer benchmarking tools, which provide originators a granular view into the real-time trends of nearly half the U.S. residential mortgage market. We also provide a Data as a Service, or DaaS, offering through private data clouds for lenders to access their own data and origination information. Revenues related to our data products are largely subscription-based and recurring in nature.
- Other:** Other revenues decreased 31% and 32% during the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, due to lower professional services related to fewer new customer implementations. Other revenues include professional services fees, as well as revenues from ancillary products. Other revenues can be both recurring and transaction-based in nature.

Operating Expenses, Operating Income/(Loss) and Operating Margin

The following chart summarizes our Mortgage Technology segment's operating expenses, operating income/(loss) and operating margin (dollars in millions). See “—Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses.

Mortgage Technology Segment:	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change*	2023	2022	Change
Operating expenses	\$ 530	\$ 557	(5)%	\$ 266	\$ 303	(13)%
Adjusted operating expenses ⁽¹⁾	\$ 301	\$ 317	(5)%	\$ 150	\$ 159	(7)%
Operating income/(loss)	\$ (45)	\$ 47	n/a	\$ (17)	\$ (6)	(158)%
Adjusted operating income ⁽¹⁾	\$ 184	\$ 287	(36)%	\$ 99	\$ 138	(28)%
Operating margin	(9) %	8 %	(17 pts)	(7) %	(2) %	(5 pts)
Adjusted operating margin ⁽¹⁾	38 %	47 %	(9 pts)	40 %	46 %	(6 pts)

*Percentage changes in the table above deemed "n/a" are not meaningful.

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. These adjusted numbers are not calculated in accordance with GAAP. See “—Non-GAAP Financial Measures”

Consolidated Operating Expenses

The following presents our consolidated operating expenses (dollars in millions):



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	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Compensation and benefits	\$ 703	\$ 714	(2)%	\$ 351	\$ 355	(1)%
Professional services	57	69	(17)	29	35	(16)
Acquisition-related transaction and integration costs	46	62	(26)	25	53	(53)
Technology and communication	345	344	—	173	169	2
Rent and occupancy	45	41	9	25	20	21
Selling, general and administrative	137	112	22	63	57	11
Depreciation and amortization	527	510	3	267	256	4
Total operating expenses	\$ 1,860	\$ 1,852	— %	\$ 933	\$ 945	(1)%

The majority of our operating expenses do not vary directly with changes in our volume and revenues, except for certain technology and communication expenses, including data acquisition costs, licensing and other fee-related arrangements and a portion of our compensation expense that is tied directly to our data sales or overall financial performance.

We expect our operating expenses to increase in absolute terms in future periods in connection with the growth of our business, and to vary from year-to-year based on the type and level of our acquisitions, integration of acquisitions and other investments.

For the six months ended June 30, 2023 and 2022, 9% and 10%, respectively, of our operating expenses were billed in pounds sterling or euros, and for the three months ended June 30, 2023 and 2022, 9% and 10%, respectively, of our operating expenses were billed in pounds sterling or euros. Due to fluctuations in the U.S. dollar compared to the pound

sterling and euro, our consolidated operating expenses were lower by \$8 million for the six months ended June 30, 2023, and were flat for the three months ended June 30, 2023, than in the comparable periods in 2022. See Item 3 “—Quantitative and Qualitative Disclosures About Market Risk —Foreign Currency Exchange Rate Risk” below for additional information.

Compensation and Benefits Expenses

Compensation and benefits expense is our most significant operating expense and includes non-capitalized employee wages, bonuses, non-cash or stock compensation, certain severance costs, benefits and employer taxes. The bonus component of our compensation and benefits expense is based on both our financial performance and individual employee performance. The performance-based restricted stock compensation expense is also based on our financial performance. Therefore, our compensation and benefits expense will vary year-to-year based on our financial performance and fluctuations in our number of employees. The below chart summarizes the significant drivers of our compensation and benefits expense results for the periods presented (dollars in millions, except employee headcount).

	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change	2023	2022	Change
Employee headcount	9,116	8,936	2 %			
Stock-based compensation expenses	\$ 83	\$ 73	14 %	\$ 43	\$ 36	18 %

Compensation and benefits expense decreased \$11 million and \$4 million for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, primarily due to lower payroll and employee benefit expenses and higher capitalized labor, partially offset by an increase in our bonus and noncash performance-based restricted stock compensation accruals and annual merit increases. The stock-based compensation expenses in the table above relate to employee stock option and restricted stock awards and exclude stock-based compensation related to acquisition-related transaction and integration costs.

Professional Services Expenses

Professional services expense includes fees for consulting services received on strategic and technology initiatives, temporary labor, as well as regulatory, legal and accounting fees, and may fluctuate as a result of changes in our use of these services in our business.

Professional services expenses decreased \$12 million and \$6 million for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, primarily due to lower consulting expenses related to bringing certain mortgage technology-related costs in-house.

Acquisition-Related Transaction and Integration Costs

We incurred \$46 million and \$25 million in acquisition-related transaction and integration costs during the six months and three months ended June 30, 2023, respectively, primarily due to legal and consulting expenses related to our pending acquisition of Black Knight and our integration of Ellie Mae, Inc., or Ellie Mae. We incurred \$62 million and \$53 million in acquisition-related transaction costs during the six months and three months ended June 30, 2022, respectively, primarily due to legal and consulting expenses related to our pending acquisition of Black Knight and our integration of Ellie Mae.

We expect to continue to explore and pursue various potential acquisitions and other strategic opportunities to strengthen our competitive position and support our growth. As a result, we may incur acquisition-related transaction costs in future periods.

Technology and Communication Expenses

Technology support services consist of costs for running our wholly-owned data centers, hosting costs paid to third-party data centers and maintenance of our computer hardware and software required to support our technology and cybersecurity. These costs are driven by system capacity, functionality and redundancy requirements. Communication expenses consist of costs or network connections for our electronic platforms and telecommunications costs.

Technology and communications expense also includes fees paid for access to external market data, licensing and other fee agreement expenses. Technology and communications expenses may be impacted by growth in electronic contract volume, our capacity requirements, changes in the number of telecommunications hubs and connections with customers to access our electronic platforms directly.

Technology and communications expenses increased \$1 million and \$4 million for the six months and three months ended June 30, 2023 respectively, from the comparable periods in 2022, primarily due to an increase in hardware and software support costs which was partially offset by a decrease in license expense.

Rent and Occupancy Expenses

Rent and occupancy expense relates to leased and owned property and includes rent, maintenance, real estate taxes, utilities and other related costs. We have significant operations located in the U.S., U.K., and India, with smaller offices located throughout the world.

Rent and occupancy expenses increased \$4 million and \$5 million for the six months and three months ended June 30, 2023, respectively, from the comparable periods in 2022, primarily due to higher costs related to rent, lease termination charges, utilities, repairs and maintenance as more employees returned to the office.

Selling, General and Administrative Expenses

Selling, general and administrative expenses include marketing, advertising, public relations, insurance, bank service charges, dues and subscriptions, travel and entertainment, non-income taxes and other general and administrative costs.

Selling, general and administrative expenses increased \$25 million for the six months ended June 30, 2023, from the comparable period in 2022 primarily due to payments of claims made following a NYSE system outage that occurred in January 2023, accruals related to potential regulatory settlements of \$11 million and increased bad debt expenses, partially offset by lower marketing costs.

Selling, general and administrative expenses increased \$6 million for the three months ended June 30, 2023, from the comparable period in 2022 primarily due to an increase in bad debt expense. This was partially offset by lower travel and entertainment expenses and lower marketing costs.

Depreciation and Amortization Expenses

Depreciation and amortization expense results from depreciation of long-lived assets such as buildings, leasehold improvements, aircraft, hardware and networking equipment, software, furniture, fixtures and equipment over their estimated useful lives. This expense includes amortization of intangible assets obtained in our acquisitions of businesses, as well as on various licensing agreements, over their estimated useful lives. Intangible assets subject to amortization consist primarily of customer relationships, trading products with finite lives and technology. This expense also includes amortization of internally-developed and purchased software over its estimated useful life.

We recorded amortization expenses on intangible assets acquired as part of our acquisitions, as well as on other intangible assets, of \$301 million and \$306 million for the six months ended June 30, 2023 and 2022, respectively and \$151 million and \$153 million for the three months ended June 30, 2023 and 2022, respectively.

We recorded depreciation expenses on our fixed assets of \$226 million and \$204 million for the six months ended June 30, 2023 and 2022, respectively, and \$116 million and \$103 million for the three months ended June 30, 2023 and 2022, respectively. This increase was primarily due to an increase in internally developed software assets in our Mortgage Technology segment.

Consolidated Non-Operating Income/(Expense)

Income and expenses incurred through activities outside of our core operations are considered non-operating. The following tables present our non-operating income/(expenses) (dollars in millions):

	Six Months Ended June 30,			Three Months Ended June 30,		
	2023	2022	Change*	2023	2022	Change
Other income/(expense):						
Interest income	\$ 193	\$ 9	n/a	\$ 102	\$ 8	n/a
Interest expense	(351)	(264)	33	(175)	(161)	8
Other expense, net	(70)	(35)	101	(35)	23	n/a
Total other income/(expense), net	\$ (228)	\$ (290)	(21)%	\$ (108)	\$ (130)	(16)%
Net income attributable to non-controlling interest	\$ (35)	\$ (21)	65%	\$ (16)	\$ (11)	53%

*Percentage changes in the table above deemed "n/a" are not meaningful.

Interest Income

Interest income increased during the six months and three months ended June 30, 2023 from the same periods in 2022 primarily due to an increase in short-term interest rates combined with larger investment balances. Interest income primarily represents interest income on our short-term investments, and for the six months and three months ended June 30, 2023, included \$124 million and \$62 million, respectively, in interest income recorded in connection with the short-term investments related to the \$5.0 billion of the SMR Notes (as defined in "Liquidity and Capital Resources—Debt") for the Black Knight acquisition. The remainder of the increase primarily relates to interest on the restricted cash and restricted investment balances held within our regulated entities.

Interest Expense

We recognized interest expense of \$351 million and \$264 million during the six months ended June 30, 2023 and 2022, respectively, and \$175 million and \$161 million during the three months ended June 30, 2023 and 2022, respectively. Interest expense for the six months and three months ended June 30, 2023 primarily represents interest expense on our outstanding debt. Interest expense increased during the six months and three months ended June 30, 2023, as we recognized \$112 million and \$56 million, respectively, additional interest expense on the \$4.9 billion of net proceeds from the SMR Notes held for the Black Knight acquisition following our May 2022 debt refinancing. See "-Debt" below.

Other Expense, net

Our equity method investments include OCC and Bakkt, among others. We recognized \$65 million and \$57 million during the six months ended June 30, 2023 and 2022, respectively, and \$30 million and \$15 million during the three months ended June 30, 2023 and 2022, respectively, of our share of estimated equity method investment losses, net, which is included in other expense, net. The estimated losses for both the six months and three months ended June 30, 2023 and June 30, 2022 are primarily related to our investment in Bakkt, partially offset by our share of net profits of OCC. Both the six month periods ended June 30, 2023 and 2022 include adjustments to reflect the difference between reported prior period actual results from our original estimates.

During the six months ended June 30, 2022, we recorded \$9 million for a legal settlement, which is included in other expense.

On May 20, 2022, we completed the sale of our Euroclear stake. The carrying value of our investment was \$700 million at the time of the sale. We recorded a net gain of \$41 million on the sale, which is included in other income during the six months and three months ended June 30, 2022.

We incurred foreign currency transaction losses of \$5 million and \$9 million for the six months ended June 30, 2023 and 2022, respectively, and \$4 million for both the three months ended June 30, 2023 and 2022, primarily attributable to the fluctuations of the pound sterling and euro relative to the U.S. dollar. Foreign currency transaction gains/(losses) are recorded in other income/(expense), net, when the settlement of foreign currency assets, liabilities and payables occur in non-functional currencies and there is an increase or decrease in the period-end foreign currency exchange rates between periods. See Item 3 "—Quantitative and Qualitative Disclosures About Market Risk—Foreign Currency Exchange Rate Risk" included elsewhere in this Quarterly Report for more information on these items.

Non-controlling Interest

For consolidated subsidiaries in which our ownership is less than 100%, and for which we have control over the assets, liabilities and management of the entity, the outside stockholders' interests are shown as non-controlling interests. As of June 30, 2023, our non-controlling interests included those related to the non-ICE limited partners' interest in our CDS clearing subsidiaries, and non-controlling interests in ICE Futures Abu Dhabi.

Consolidated Income Tax Provision

Consolidated income tax expense was \$207 million and \$338 million for the six months ended June 30, 2023 and 2022, respectively, and \$32 million and \$173 million for the three months ended June 30, 2023 and June 30, 2022, respectively. The change in consolidated income tax expense between periods is primarily due to the tax impact of changes in our pre-tax income and the changes in our effective tax rate each period.

Our effective tax rate was 12% and 22% for the six months ended June 30, 2023 and 2022, respectively, and 4% and 23% for the three months ended June 30, 2023 and June 30, 2022, respectively. The effective tax rates for the six months and three months ended June 30, 2023 were lower than the effective tax rates for the comparable periods in 2022 primarily due to the tax benefits resulting from favorable audit settlements for prior years, deferred and current tax benefits from state apportionment changes and the application of the high-tax exception to the Global Intangible Low-Taxed Income, or GILTI, in the current period, partially offset by the impact of the U.K. corporate income tax increase from 19% to 25% effective April 1, 2023.

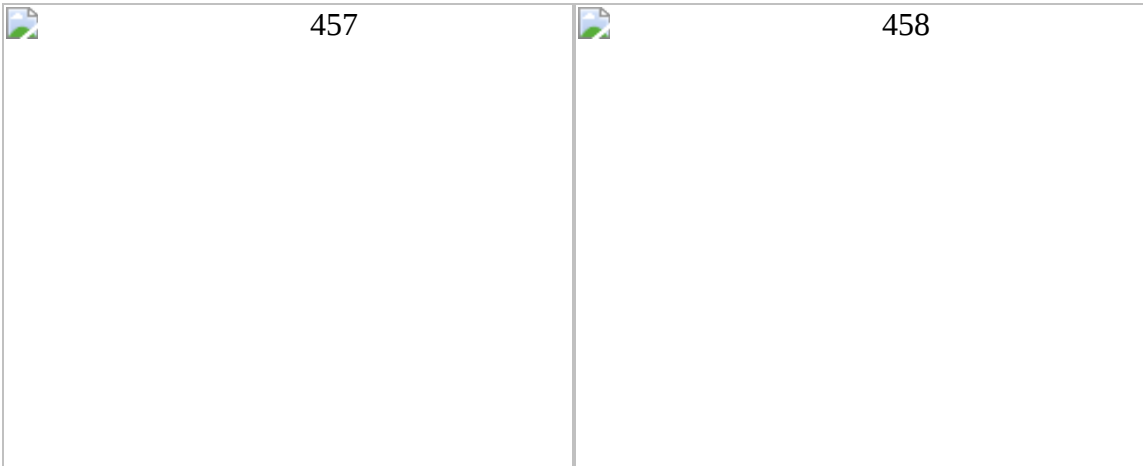
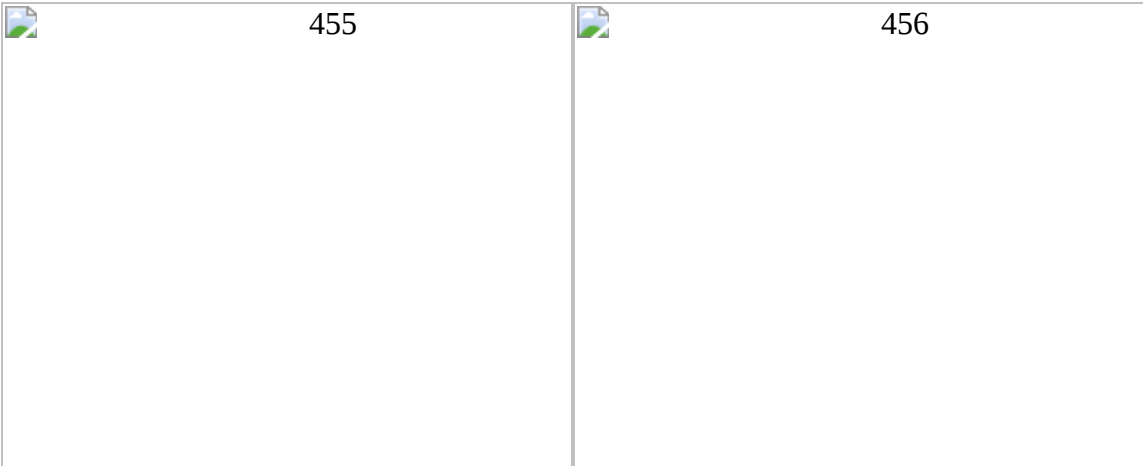
In conjunction with the increase in the U.K. corporate income tax rate, we intend to elect the GILTI high-tax exception election in 2023. During the three months ended June 30, 2023, our tax provision includes the impacts of this election, which resulted in a reduction in our GILTI inclusions. Our unrecognized tax benefit as of June 30, 2023 was \$240 million, a \$7 million net decrease from the \$247 million as of December 31, 2022. The net decrease includes a \$40 million reduction as a result of audit settlements, a \$20 million increase related to current year positions, a \$31 million increase related to prior year positions, and an \$18 million reduction related to prior year positions.

In August 2022, the Inflation Reduction Act of 2022, or IRA, was signed into law. The IRA introduced a 15% corporation minimum tax, or CAMT, on adjusted financial statement income for corporations with profits in excess of \$1 billion, effective for tax years after December 31, 2022. Based on the current guidance provided by the Internal Revenue Service and Department of the Treasury, the implementation of the CAMT did not have a material impact on our financial statements as of June 30, 2023.

The IRA also includes a share buyback excise tax of 1% on share repurchases, which will apply to net share repurchases after December 31, 2022. During the six months ended June 30, 2023, we did not repurchase any shares, therefore, we were not subject to any excise tax. The newly imposed excise tax on share repurchases is not considered an income tax. Any excise tax, as a result of future share repurchases, will be considered part of the cost of the shares repurchased and reflected in the equity section of our consolidated financial statements.

Liquidity and Capital Resources

Below are charts that reflect our outstanding debt and capital allocation. The acquisition and integration costs in the chart below include cash paid for acquisitions, net of cash received for divestitures, cash paid for equity and equity method investments, cash paid for non-controlling interest and redeemable non-controlling interest, and acquisition-related transaction and integration costs in each period.



We have financed our operations, growth and cash needs primarily through income from operations and borrowings under our various debt facilities. Our principal capital requirements have been to fund capital expenditures, working capital, strategic acquisitions and investments, stock repurchases, dividends and the development of our technology platforms. We believe that our cash on hand and cash flows from operations will be sufficient to repay our outstanding debt, but we

may also need to incur additional debt or issue additional equity securities in the future. See “- Future Capital Requirements” below.

See “– Cash Flow” below for a discussion of our capital expenditures and capitalized software development costs.

Consolidated cash and cash equivalents were \$2.9 billion and \$1.8 billion as of June 30, 2023 and December 31, 2022, respectively. We had \$5.6 billion and \$6.6 billion in short-term and long-term restricted cash and cash equivalents as of June 30, 2023 and December 31, 2022, respectively. We had \$86.9 billion and \$142.0 billion of cash and cash equivalent margin deposits and guaranty funds as of June 30, 2023 and December 31, 2022, respectively.

As of June 30, 2023, the amount of unrestricted cash held by our non-U.S. subsidiaries was \$460 million. Due to the application of GILTI as of January 1, 2018, the majority of our foreign earnings as of December 31, 2022 have been subject to immediate U.S. income taxation and can be distributed to the U.S. in the future with no material additional income tax consequences. As we decided to make the GILTI high-tax exception election in the current period, the majority of our foreign earnings in 2023 are not expected to be subject to immediate U.S. income taxation; however, these foreign earnings can also generally be distributed to the U.S. with no additional income tax consequences.

Our cash and cash equivalents and financial investments are managed as a global treasury portfolio of non-speculative financial instruments that are readily convertible into cash, such as overnight deposits, term deposits, money market funds, mutual funds for treasury investments, short duration fixed income investments and other money market instruments, thus ensuring high liquidity of financial assets. We may invest a portion of our cash in excess of short-term operating needs in investment-grade marketable debt securities, including government or government-sponsored agencies and corporate debt securities.

Cash Flow

The following table presents the major components of net changes in cash and cash equivalents, and restricted cash and cash equivalents (in millions):

	Six Months Ended June 30,	
	2023	2022
Net cash provided by/(used in):		
Operating activities	\$ 1,805	\$ 1,725
Investing activities	1,144	2,843
Financing activities	(57,886)	19,238
Effect of exchange rate changes	6	(19)
Net increase/(decrease) in cash and cash equivalents, restricted cash and cash equivalents, and cash and cash equivalent margin deposits and guaranty funds	\$ (54,931)	\$ 23,787

Operating Activities

Net cash provided by operating activities primarily consists of net income adjusted for certain items, including depreciation and amortization, deferred taxes, stock based compensation and the effects of changes in working capital.

The \$80 million increase in net cash provided by operating activities during the six months ended June 30, 2023 from the comparable period in 2022 was primarily driven by an increase in net income, and to a lesser extent, the gain on our sale of our Euroclear investment during the six months ended June 30, 2022. The remaining fluctuations are due to changes in our working capital and the timing of various payments, such as lower Section 31 fee payments of \$165 million.

Investing Activities

Consolidated net cash provided by investing activities for the six months ended June 30, 2023 primarily relates to \$3.1 billion in proceeds from the sale of invested margin deposits, partially offset by \$949 million of purchases of investments, \$771 million of purchases of invested margin deposits, \$61 million of capital expenditures and \$142 million of capitalized software development costs.

Consolidated net cash provided by investing activities for the six months ended June 30, 2022 primarily relates to \$3.8 billion in proceeds from the sale of invested margin deposits and \$741 million in proceeds from the sale of our Euroclear investment, partially offset by \$1.4 billion of purchases of invested margin deposits, \$70 million of capital expenditures and \$134 million of capitalized software development costs.

The capital expenditures primarily relate to hardware and software purchases to continue the development and expansion of our electronic platforms, data services and clearing houses, and leasehold improvements. The software development

expenditures primarily relate to the development and expansion of our electronic trading platforms, data services, mortgage services and clearing houses.

Financing Activities

Consolidated net cash used in financing activities for the six months ended June 30, 2023 primarily relates to a decrease in our cash and cash equivalent margin deposits and guaranty fund balances of \$57.4 billion due to lower commodity prices and reduced volatility, \$472 million in dividend payments to stockholders and \$50 million in cash payments related to treasury shares received for restricted stock tax payments and stock option exercises.

Consolidated net cash provided by financing activities for the six months ended June 30, 2022 primarily relates to an increase in our cash and cash equivalent margin deposits and guaranty fund balances of \$16.2 billion due to increased volatility and \$5.2 billion in net proceeds from our debt facilities, partially offset by \$1.0 billion in net repayments under our Commercial Paper Program, \$632 million in repurchases of our common stock, \$427 million in dividend payments to stockholders and \$71 million in cash payments related to treasury shares received for restricted stock tax payments and stock option exercises.

Debt

As of June 30, 2023, we had \$18.1 billion in outstanding debt, all of which related to our senior notes, with a weighted average maturity of 16 years and a weighted average cost of 3.6% per annum. We did not have any commercial paper notes outstanding as of June 30, 2023. As of December 31, 2022, we had \$18.1 billion in outstanding debt, all of which related to our senior notes. We also had \$4 million outstanding under credit lines at our ICE India subsidiaries. As of December 31, 2022, our senior notes of \$18.1 billion had a weighted average maturity of 16 years and a weighted average cost of 3.6% per annum. We did not have any commercial paper notes outstanding as of December 31, 2022.

We have a \$3.9 billion senior unsecured revolving credit facility, or the Credit Facility, with a maturity date of May 25, 2027. As of June 30, 2023, of the \$3.9 billion that was available for borrowing under the Credit Facility, \$171 million was required to support certain broker-dealer and other subsidiary commitments. The remaining \$3.7 billion was available for working capital and general corporate purposes including, but not limited to, acting as a backstop to future increases in the amounts outstanding under the Commercial Paper Program.

We intend to use \$4.9 billion net proceeds of our senior notes due in 2025, 2027, 2029 and 2062, or collectively, the SMR Notes, together with the issuance of commercial paper and/or borrowings under the Credit Facility, cash on hand or other immediately available funds and borrowings under the Term Loan, to finance the cash portion of the purchase price for Black Knight. The SMR Notes are subject to a special mandatory redemption feature pursuant to which we will be required to redeem all of the outstanding SMR Notes at a redemption price equal to 101% of the aggregate principal amount of the SMR Notes, plus accrued and unpaid interest, in the event that the Black Knight acquisition is not consummated on or prior to May 4, 2023 (subject to two automatic extensions of three months each, to August 4, 2023 and to November 4, 2023, respectively, if U.S. antitrust clearance or a related law, injunction, order or other judgment, in each case whether temporary, preliminary or permanent, that restrains, enjoins or otherwise prohibits the consummation of the Black Knight merger remains outstanding and all other conditions to closing are satisfied (or in the case of conditions that by their terms are to be satisfied at the closing, are capable of being satisfied if the closing were to occur on such date) at each extension date), or if the Black Knight merger agreement is terminated at any time prior to such date. The \$4.9 billion net proceeds from the SMR Notes are separately invested and recorded as short-term restricted cash and cash equivalents in our consolidated balance sheet as of June 30, 2023. For additional information regarding this transaction, refer to Note 3 to our consolidated unaudited financial statements, included in this Quarterly Report.

We have a \$2.4 billion two-year senior unsecured delayed draw term loan facility, or the Term Loan. Draws under the Term Loan bear interest on the principal amount outstanding at either (a) Term SOFR plus an applicable margin plus a credit spread adjustment of 10 basis points or (b) a "base rate" plus an applicable margin. The applicable margin ranges from 0.625% to 1.125% for Term SOFR loans and from 0.000% to 0.125% for base rate loans, in each case, based on a ratings-based pricing grid. We expect to use the proceeds from borrowings under the Term Loan to fund a portion of the purchase price for the Black Knight acquisition. We have the option to prepay outstanding amounts under the Term Loan in whole or in part at any time. No amounts were outstanding under the Term Loan as of June 30, 2023.

Our Commercial Paper Program enables us to borrow efficiently at reasonable short-term interest rates and provides us with the flexibility to de-lever using our strong annual cash flows from operating activities whenever our leverage becomes elevated as a result of investment or acquisition activities. We did not have any notes outstanding under our Commercial Paper Program as of June 30, 2023.

Upon maturity of our commercial paper and to the extent old issuances are not repaid by cash on hand, we are exposed to the rollover risk of not being able to issue new commercial paper. To mitigate this risk, we maintain the Credit Facility for

an aggregate amount which meets or exceeds the amount issued under our Commercial Paper Program at any time. If we were not able to issue new commercial paper, we have the option of drawing on the backstop revolving facility. However, electing to do so would result in higher interest expense.

For additional details of our debt instruments, refer to Note 8 to our consolidated unaudited financial statements, included in this Quarterly Report, and Note 10 to our consolidated financial statements included in our 2022 Form 10-K.

Capital Return

In December 2021, our Board approved an aggregate of \$3.15 billion for future repurchases of our common stock with no fixed expiration date that became effective January 1, 2022.

We did not have any share repurchases during the six months ended June 30, 2023. For the six months ended June 30, 2022, we repurchased 5.0 million shares of our outstanding common stock at a cost of \$632 million, including 4.6 million shares at a cost of \$582 million under our Rule 10b5-1 trading plan and 0.4 million shares at a cost of \$50 million on the open market during an open trading period. For the three months ended June 30, 2022, we repurchased 1.3 million shares of our outstanding common stock at a cost of \$157 million under our Rule 10b5-1 trading plan. Shares repurchased are held in treasury stock.

The remaining balance of Board approved funds for future repurchases as of June 30, 2023 is \$2.5 billion. In connection with our pending acquisition of Black Knight, on May 4, 2022 we terminated our Rule 10b5-1 trading plan and suspended share repurchases. The approval of our Board for stock repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board may increase or decrease the amount available for repurchases from time to time.

From time to time, we enter into Rule 10b5-1 trading plans, as authorized by our Board, to govern some or all of the repurchases of our shares of common stock. We may discontinue stock repurchases at any time and may amend or terminate a Rule 10b5-1 trading plan at any time, subject to applicable rules. We expect funding for any stock repurchases to come from our operating cash flow or borrowings under our Commercial Paper Program or our debt facilities. The timing and extent of future repurchases that are not made pursuant to a Rule 10b5-1 trading plan will be at our discretion and will depend upon many conditions. In making a determination regarding any stock repurchases, management considers multiple factors, including overall stock market conditions, our common stock price performance, the remaining amount authorized for repurchases by our Board, the potential impact of a stock repurchase program on our corporate debt ratings, our expected free cash flow and working capital needs, our current and future planned strategic growth initiatives, and other potential uses of our cash and capital resources.

During the three months ended June 30, 2023, we paid a quarterly dividend of \$0.42 per share of our common stock for an aggregate payout of \$236 million, which includes the payment of dividend equivalents on unvested employee restricted stock units.

Future Capital Requirements

Our future capital requirements will depend on many factors, including the rate of growth across our segments, strategic plans and acquisitions, available sources for financing activities, required and discretionary technology and clearing initiatives, regulatory requirements, the timing and introduction of new products and enhancements to existing products, the geographic mix of our business and potential stock repurchases.

We currently expect to incur capital expenditures (including operational and real estate capital expenditures) and to incur software development costs that are eligible for capitalization ranging in the aggregate between \$450 million and \$500 million in 2023, which we believe will support the enhancement of our technology, business integration and the continued growth of our businesses.

As of June 30, 2023, we had \$2.5 billion authorized for future repurchases of our common stock. Refer to Note 10 to our consolidated financial statements included in this Quarterly Report for additional details on our stock repurchase program.

Our Board has adopted a quarterly dividend policy providing that dividends will be approved quarterly by the Board or the Audit Committee taking into account factors such as our evolving business model, prevailing business conditions, our current and future planned strategic growth initiatives and our financial results and capital requirements, without a predetermined net income payout ratio. On August 3, 2023, we announced a \$0.42 per share dividend for the third quarter of 2023 with the dividend payable on September 29, 2023 to stockholders of record as of September 15, 2023.

Other than the facilities for the ICE Clearing Houses, our Credit Facility, our Term Loan, our Bridge Facility and our Commercial Paper Program are currently the only significant agreements or arrangements that we have for liquidity and capital resources with third parties. See Notes 8 and 12 to our consolidated financial statements included in this Quarterly Report for further discussion. In the event of any strategic acquisitions, mergers or investments, or if we are required to raise capital for any reason or desire to return capital to our stockholders, we may incur additional debt, issue additional equity to raise necessary funds, repurchase additional shares of our common stock or pay a dividend. However, we cannot provide assurance that such financing or transactions will be available or successful, or that the terms of such financing or transactions will be favorable to us. See “—Debt” above.

Non-GAAP Measures

Non-GAAP Financial Measures

We use certain financial measures internally to evaluate our performance and make financial and operational decisions that are presented in a manner that adjusts from their equivalent GAAP measures or that supplement the information provided by our GAAP measures. We use these adjusted results because we believe they more clearly highlight trends in our business that may not otherwise be apparent when relying solely on GAAP financial measures, since these measures eliminate from our results specific financial items that have less bearing on our core operating performance.

We use these measures in communicating certain aspects of our results and performance, including in this Quarterly Report, and believe that these measures, when viewed in conjunction with our GAAP results and the accompanying reconciliation, can provide investors with greater transparency and a greater understanding of factors affecting our financial condition and results of operations than GAAP measures alone. In addition, we believe the presentation of these measures is useful to investors for making period-to-period comparisons of results because the adjustments to GAAP are not reflective of our core business performance.

These financial measures are not presented in accordance with, or as an alternative to, GAAP financial measures and may be different from non-GAAP measures used by other companies. We encourage investors to review the GAAP financial measures included in this Quarterly Report, including our consolidated financial statements, to aid in their analysis and understanding of our performance and in making comparisons.

The tables below outline our adjusted operating expenses, adjusted operating income, adjusted operating margin, adjusted net income attributable to ICE common stockholders, adjusted diluted earnings per share and adjusted free cash flow, which are non-GAAP measures that are calculated by making adjustments for items we view as not reflective of our cash operations and core business performance. These measures, including the adjustments and their related income tax effect and other tax adjustments (in millions, except for percentages and per share amounts), are as follows:

	Exchanges Segment		Fixed Income and Data Services Segment		Mortgage Technology Segment		Consolidated		
	Six Months Ended June 30,								
Operating income adjustments:	2023	2022	2023	2022	2023	2022	2023	2022	
Total revenues, less transaction-based expenses	\$ 2,190	\$ 2,088	\$ 1,109	\$ 1,021	\$ 485	\$ 604	\$ 3,784	\$ 3,713	
Operating expenses	631	603	699	692	530	557	1,860	1,852	
Less: Amortization of acquisition-related intangibles	33	33	85	93	183	180	301	306	
Less: Transaction and integration costs	—	—	—	—	46	60	46	60	
Less: Other	17	—	—	—	—	—	17	—	
Adjusted operating expenses	\$ 581	\$ 570	\$ 614	\$ 599	\$ 301	\$ 317	\$ 1,496	\$ 1,486	
Operating income/(loss)	\$ 1,559	\$ 1,485	\$ 410	\$ 329	\$ (45)	\$ 47	\$ 1,924	\$ 1,861	
Adjusted operating income	\$ 1,609	\$ 1,518	\$ 495	\$ 422	\$ 184	\$ 287	\$ 2,288	\$ 2,227	
Operating margin	71 %	71 %	37 %	32 %	(9)%	8 %	51 %	50 %	
Adjusted operating margin	73 %	73 %	45 %	41 %	38 %	47 %	60 %	60 %	
Non-operating income adjustments:									
Net income attributable to ICE common stockholders							\$ 1,454	\$ 1,212	
Add: Amortization of acquisition-related intangibles							301	306	
Add: Transaction and integration costs							46	60	
Add/(Less): Net interest (income)/expense on pre-acquisition-related debt and debt extinguishment							(12)	48	
Less: Gain on sale of Euroclear equity investment and dividends received							—	(41)	
Add: Net losses from unconsolidated investees							65	57	
Add: Other							17	9	
Less: Income tax effect for the above items							(112)	(123)	
Add/(Less): Deferred tax adjustments on acquisition-related intangibles							(85)	15	
Less: Other tax adjustments							(81)	—	
Adjusted net income attributable to ICE common stockholders							\$ 1,593	\$ 1,543	
Diluted earnings per share attributable to ICE common stockholders							\$ 2.59	\$ 2.16	
Adjusted diluted earnings per share attributable to ICE common stockholders							\$ 2.84	\$ 2.75	
Diluted weighted average common shares outstanding							561	562	

	Exchanges Segment		Fixed Income and Data Services Segment		Mortgage Technology Segment		Consolidated	
	Three Months Ended June 30,							
Operating income adjustments:	2023	2022	2023	2022	2023	2022	2023	2022
Total revenues, less transaction-based expenses	\$ 1,093	\$ 1,005	\$ 546	\$ 512	\$ 249	\$ 297	\$ 1,888	\$ 1,814
Operating expenses	311	304	356	338	266	303	933	945
Less: Amortization of acquisition-related intangibles	17	17	43	44	91	92	151	153
Less: Transaction and integration costs	—	—	—	—	25	52	25	52
Less: Other	1	—	—	—	—	—	1	—
Adjusted operating expenses	\$ 293	\$ 287	\$ 313	\$ 294	\$ 150	\$ 159	\$ 756	\$ 740
Operating income/(loss)	\$ 782	\$ 701	\$ 190	\$ 174	\$ (17)	\$ (6)	\$ 955	\$ 869
Adjusted operating income	\$ 800	\$ 718	\$ 233	\$ 218	\$ 99	\$ 138	\$ 1,132	\$ 1,074
Operating margin	72 %	70 %	35 %	34 %	(7)%	(2)%	51 %	48 %
Adjusted operating margin	73 %	71 %	43 %	43 %	40 %	46 %	60 %	59 %
Non-operating income adjustments:								
Net income attributable to ICE common stockholders							\$ 799	\$ 555
Add: Amortization of acquisition-related intangibles							151	153
Add: Transaction and integration costs							25	52
Add/(Less): Net interest (income)/expense on pre-acquisition-related debt and debt extinguishment							(6)	48
Less: Gain on sale of Euroclear equity investment and dividends received							—	(41)
Add: Net losses from unconsolidated investees							30	15
Add: Other							1	—
Less: Income tax effect for the above items							(55)	(65)
Add/(Less): Deferred tax adjustments on acquisition-related intangibles							(86)	22
Less: Other tax adjustments							(57)	—
Adjusted net income attributable to ICE common stockholders							\$ 802	\$ 739
Diluted earnings per share attributable to ICE common stockholders							\$ 1.42	\$ 0.99
Adjusted diluted earnings per share attributable to ICE common stockholders							\$ 1.43	\$ 1.32
Diluted weighted average common shares outstanding							561	560

Amortization of acquisition-related intangibles are included in non-GAAP adjustments as excluding these non-cash expenses provides greater clarity regarding our financial strength and stability of cash operating results.

Transaction and integration costs are included as part of our core business expenses, except for those that are directly related to the announcement, closing, financing, or termination of a transaction. However, we adjust for the acquisition-related transaction and integration costs for acquisitions such as Ellie Mae given the magnitude of the \$11.4 billion purchase price of the acquisition. We also adjusted for the acquisition-related transaction costs related to our pending acquisition of Black Knight.

We adjust for our share of net income/(losses) related to our equity method investments, which primarily include OCC and Bakkt. We believe these adjustments provide greater clarity of our performance, given that equity and equity method investments are non-cash and not a part of our core operations.

Other non-GAAP adjustments during the six months ended June 30, 2023 relate to a \$6 million expense for claims made following a NYSE system outage that occurred in January 2023 and accruals related to potential regulatory settlements of \$11 million. Other non-GAAP adjustments during the six months ended June 30, 2022 relate to an accrual for a legal settlement. During the three months ended June 30, 2022 we exclude the \$30 million of costs associated with the May and June 2022 extinguishment of four series of senior notes that would have matured in 2022 and 2023 using proceeds from our May 2022 issuance of new senior notes as a non-GAAP adjustment. We also exclude \$23 million of interest expense on pre-acquisition-related debt from our May 2022 debt refinancing related to the pending Black Knight acquisition. This adjustment is net of \$5 million of interest income earned on investments from the pre-acquisition debt proceeds. Additionally, we exclude the gain on the sale of our Euroclear investment and the related historical dividends received. We do not consider events of this type to be reflective of our core business.

Non-GAAP tax adjustments include the tax impacts of the pre-tax non-GAAP adjustments, deferred tax adjustments on acquisition-related intangibles, and other tax adjustments. The deferred tax adjustments of (\$85 million) and (\$86 million) for the six months and three months ended June 30, 2023, respectively, relate primarily to U.S. state apportionment changes. The deferred tax adjustments of \$15 million and \$22 million for the six months and three months ended June 30, 2022, respectively, relate primarily to U.S. state apportionment changes. Other tax adjustments of (\$81 million) and

(\$57 million) during the six months and three months ended June 30, 2023, respectively, are primarily related to audit settlements for pre-acquisition tax matters as well as state apportionment changes in prior years.

Non-GAAP Liquidity Measures

We consider adjusted free cash flow to be a non-GAAP liquidity measure that provides useful information to management and investors to analyze cash resources generated from our operations. We believe that free cash flow is also useful as one of the bases for comparing our performance with our competitors, and demonstrates our ability to convert the reinvestment of capital expenditures and capitalized software development costs required to maintain and grow our business, as well as adjust for timing differences related to the payment of section 31 fees. This non-GAAP liquidity measure is not presented in accordance with, or as an alternative to, GAAP liquidity measures and may be different from non-GAAP measures used by other companies. Adjusted free cash flow, including the related adjustments are as follows (in millions):

	Six Months Ended June 30,	
	2023	2022
Net cash provided by operating activities	\$ 1,805	\$ 1,725
Less: Capital expenditures	(61)	(70)
Less: Capitalized software development costs	(142)	(134)
Free cash flow	1,602	1,521
Add/(less): Section 31 fees, net	50	(115)
Adjusted free cash flow	\$ 1,652	\$ 1,406

For additional information on these items, refer to our consolidated financial statements included in this Quarterly Report and “— Consolidated Operating Expenses” above.

Contractual Obligations and Commercial Commitments

During the six months ended June 30, 2023, there were no significant changes to our contractual obligations and commercial commitments from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2022 Form 10-K.

As described in Note 12 to our consolidated financial statements, which are included elsewhere in this Quarterly Report, certain clearing house collateral is reported off-balance sheet. We do not have any relationships with unconsolidated entities or financial partnerships, often referred to as structured finance or special purpose entities.

New and Recently Adopted Accounting Pronouncements

During the six months ended June 30, 2023, there were no significant changes to the new and recently adopted accounting pronouncements applicable to us from those disclosed in Note 2 of our 2022 Form 10-K.

Critical Accounting Policies

During the six months ended June 30, 2023, there were no significant changes to our critical accounting policies and estimates from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2022 Form 10-K.

ITEM 3. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

As a result of our operating and financing activities, we are exposed to market risks such as interest rate risk, foreign currency exchange rate risk and credit risk. We have implemented policies and procedures designed to measure, manage, monitor and report risk exposures, which are regularly reviewed by the appropriate management and supervisory bodies.

Interest Rate Risk

We have exposure to market risk for changes in interest rates relating to our cash and cash equivalents, short-term and long-term restricted cash and cash equivalents, short-term and long-term investments and indebtedness. As of June 30,

2023 and December 31, 2022, our cash and cash equivalents and short-term and long-term restricted cash and cash equivalents were \$8.5 billion and \$8.4 billion, respectively, of which \$297 million and \$346 million, respectively, were denominated in pounds sterling, euros or Canadian dollars, and the remaining amounts are denominated in U.S. dollars. We do not use our investment portfolio for trading or other speculative purposes. A hypothetical 50% decrease in short-term interest rates would decrease our annual pre-tax earnings by \$201 million as of June 30, 2023, assuming no change in the amount or composition of our cash and cash equivalents and short-term and long-term restricted cash and cash equivalents.

As of June 30, 2023, we had \$18.1 billion in outstanding debt, all of which related to our senior notes. See Part I, Item 2 "Management's Discussion and Analysis of Financial Condition and Results of Operations—Debt" and Note 8 to our consolidated financial statements included in this Quarterly Report.

The interest rates on our Commercial Paper Program are currently evaluated based upon current maturities and market conditions. As of June 30, 2023, we did not have any notes outstanding under our Commercial Paper Program. The effective interest rate of commercial paper issuances will fluctuate based on the movement in short-term interest rates along with shifts in supply and demand within the commercial paper market.

Foreign Currency Exchange Rate Risk

As an international business, we are subject to foreign currency exchange rate risk. We may experience gains or losses from foreign currency transactions in the future given that a significant part of our assets and liabilities are recorded in pounds sterling, Canadian dollars or euros, and a significant portion of our revenues and expenses are recorded in pounds sterling or euros. Certain assets, liabilities, revenues and expenses of foreign subsidiaries are denominated in the local functional currency of such subsidiaries. Our exposure to foreign denominated earnings for the six months and three months ended June 30, 2023 and 2022 is presented by primary foreign currency in the following table (dollars in millions, except exchange rates):

	Six Months Ended June 30, 2023		Three Months Ended June 30, 2023		Six Months Ended June 30, 2022		Three Months Ended June 30, 2022	
	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro
Average exchange rate to the U.S. dollar in the current year period	1.2336	1.0812	1.2522	1.0893	1.2996	1.0939	1.2568	1.0648
Average exchange rate to the U.S. dollar in the same period in the prior year	1.2996	1.0939	1.2568	1.0648	1.3889	1.2057	1.3985	1.2054
Average exchange rate increase/ (decrease)	(5)%	(1)%	— %	2 %	(6)%	(9)%	(10)%	(12)%
Foreign denominated percentage of:								
Revenues, less transaction-based expenses	7 %	8 %	6 %	8 %	8 %	7 %	8 %	6 %
Operating expenses	7 %	2 %	7 %	2 %	8 %	2 %	8 %	2 %
Operating income	7 %	13 %	5 %	14 %	7 %	11 %	7 %	11 %
Impact of the currency fluctuations ⁽¹⁾ on:								
Revenues, less transaction-based expenses	\$ (16)	\$ (3)	\$ (1)	\$ 4	\$ (20)	\$ (24)	\$ (16)	\$ (14)
Operating expenses	\$ (8)	\$ —	\$ (1)	\$ 1	\$ (10)	\$ (4)	\$ (8)	\$ (3)
Operating income	\$ (8)	\$ (3)	\$ —	\$ 3	\$ (10)	\$ (20)	\$ (8)	\$ (11)

(1) Represents the impact of currency fluctuation for the six months and three months ended June 30, 2023 and 2022 compared to the same periods in the prior year.

We have a significant part of our assets, liabilities, revenues and expenses recorded in pounds sterling or euros. During both the six months and three months ended June 30, 2023, 15% and 14%, respectively, of our consolidated revenues, less transaction-based expenses were denominated in pounds sterling or euros and for both the six months and three months ended June 30, 2023, 9% of our consolidated operating expenses were denominated in pounds sterling or euros. As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues and expenses denominated in foreign currencies changes accordingly.

Foreign currency transaction risk related to the settlement of foreign currency denominated assets, liabilities and payables occurs through our operations, which are received in or paid in pounds sterling, Canadian dollars, or euros, due to the increase or decrease in the foreign currency exchange rates between periods. We incurred foreign currency transaction losses of \$5 million and \$9 million for the six months ended June 30, 2023 and 2022, respectively, and \$4 million for both

the three months ended June 30, 2023 and 2022, inclusive of the impact of foreign currency hedging transactions. The foreign currency transaction losses were primarily attributable to the fluctuations of the pound sterling and euro relative to the U.S. dollar. A 10% adverse change in the underlying foreign currency exchange rates as of June 30, 2023, assuming no change in the composition of the foreign currency denominated assets, liabilities and payables and assuming no hedging activity, would result in a foreign currency loss of \$8 million.

We entered into foreign currency hedging transactions during the six months and three months ended June 30, 2023 and 2022 as economic hedges to help mitigate a portion of our foreign exchange risk exposure and may enter into additional hedging transactions in the future to help mitigate our foreign exchange risk exposure. Although we may enter into additional hedging transactions in the future, these hedging arrangements may not be effective, particularly in the event of imprecise forecasts of the levels of our non-U.S. denominated assets and liabilities.

We have foreign currency translation risk equal to our net investment in our foreign subsidiaries. The financial statements of these subsidiaries are translated into U.S. dollars using a current rate of exchange, with gains or losses included in the cumulative translation adjustment account, a component of equity. Our exposure to the net investment in foreign currencies is presented by primary foreign currencies in the table below (in millions):

	As of June 30, 2023		
	Position in pounds sterling	Position in Canadian dollars	Position in euros
Assets	£ 697	\$ 2,062	€ 184
of which goodwill represents	550	392	92
Liabilities	95	1,578	62
Net currency position	£ 602	\$ 484	€ 122
Net currency position, in \$USD	\$ 764	\$ 365	\$ 133
Negative impact on consolidated equity of a 10% decrease in foreign currency exchange rates	\$ 76	\$ 37	\$ 13

Foreign currency translation adjustments are included as a component of accumulated other comprehensive income/(loss) within our balance sheet. See the table below for the portion of equity attributable to foreign currency translation adjustments as well as the activity for the six months and three months ended June 30, 2023 included within our statement of other comprehensive income. The impact of the foreign currency exchange rate differences in the tables below were primarily driven by fluctuations of the pound sterling as compared to the U.S. dollar which were 1.2697, 1.2332 and 1.2093 as of June 30, 2023, March 31, 2023 and December 31, 2022, respectively, and by fluctuations of the euro as compared to the U.S. dollar which were 1.0911, 1.0842 and 1.0704 as of June 30, 2023, March 31, 2023 and December 31, 2022, respectively.

	Changes in Accumulated Other Comprehensive Loss from Foreign Currency Translation Adjustments (in millions)
Balance, as of December 31, 2022	\$ (278)
Other comprehensive income	47
Income tax benefit/(expense)	—
Net current period other comprehensive income	47
Balance, as of June 30, 2023	\$ (231)

	Changes in Accumulated Other Comprehensive Loss from Foreign Currency Translation Adjustments (in millions)
Balance, as of March 31, 2023	\$ (262)
Other comprehensive income	31
Income tax benefit/(expense)	—
Net current period other comprehensive income	31
Balance, as of June 30, 2023	\$ (231)

The future impact on our business relating to the U.K. leaving the EU and the corresponding regulatory changes are uncertain at this time, including future impacts on currency exchange rates.

Credit Risk

We are exposed to credit risk in our operations in the event of a counterparty default. We limit our exposure to credit risk by rigorously selecting the counterparties with which we make our investments, monitoring them on an ongoing basis and executing agreements to protect our interests.

Clearing House Cash Deposit Risks

The ICE Clearing Houses hold material amounts of clearing member margin deposits which are held or invested primarily to provide security of capital while minimizing credit, market and liquidity risks. Refer to Note 12 to our consolidated financial statements for more information on the ICE Clearing Houses' cash and cash equivalent margin deposits and guaranty funds, invested deposits, delivery contracts receivable and unsettled variation margin which were \$88.3 billion as of June 30, 2023. While we seek to achieve a reasonable rate of return which may generate interest income for our clearing members, we are primarily concerned with preservation of capital and managing the risks associated with these deposits. As the ICE Clearing Houses may pass on interest revenues (minus costs) to the clearing members, this could include negative or reduced yield due to market conditions. For a summary of the risks associated with these deposits and how these risks are mitigated, see Part II, Item 7(A) "Quantitative and Qualitative Disclosures About Market Risk" in our 2022 Form 10-K.

ITEM 4. CONTROLS AND PROCEDURES

(a) *Evaluation of Disclosure Controls and Procedures.* As of the end of the period covered by this report, an evaluation was carried out by our management, with the participation of our Chief Executive Officer and Chief Financial Officer, of the effectiveness of our disclosure controls and procedures (as defined in Rule 13a-15(e) under the Securities Exchange Act of 1934). Based upon that evaluation, our Chief Executive Officer and Chief Financial Officer concluded that these disclosure controls and procedures were effective as of the end of the period covered by this report.

(b) *Changes in Internal Controls over Financial Reporting.* There were no changes in our internal controls over financial reporting that occurred during our most recent fiscal quarter that have materially affected, or are reasonably likely to materially affect, our internal controls over financial reporting.

PART II. Other Information

ITEM 1. LEGAL PROCEEDINGS

See Note 13 to the consolidated financial statements and related notes, which is incorporated by reference herein.

ITEM 1(A). RISK FACTORS

During the six months ended June 30, 2023, there were no significant new risk factors from those disclosed in Part I, Item 1A, "Risk Factors" in our 2022 Form 10-K. In addition to the other information set forth in this Quarterly Report, including the information in the "—Regulation" section of Item 2 - Management's Discussion and Analysis of Financial Condition and Results of Operations, you should carefully consider the factors discussed under "Risk Factors" and the regulation discussion under "Business—Regulation" in our 2022 Form 10-K. These risks could materially and adversely affect our business, financial condition and results of operations. The risks and uncertainties in our 2022 Form 10-K are not the only ones facing us. Additional risks and uncertainties not presently known to us, or that we currently believe to be immaterial, may also adversely affect our business.

ITEM 2. UNREGISTERED SALES OF EQUITY SECURITIES AND USE OF PROCEEDS

Stock Repurchases

During the six months ended June 30, 2023, there were not any purchases made by or on behalf of ICE or any "affiliated purchaser" (as defined in Rule 10b-18(a)(3) under the Exchange Act) of our common stock.

Refer to Note 10 to our consolidated financial statements and related notes, which are included elsewhere in this Quarterly Report, for details on our stock repurchases.

ITEM 3. DEFAULTS UPON SENIOR SECURITIES

None.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

ITEM 5. OTHER INFORMATION

On May 17, 2023, Martha A. Tirinnanzi, one of our directors, adopted a trading plan for the sale of shares of ICE common stock, which is intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) under the Exchange Act. The plan expires on the earlier of (i) April 30, 2024 or (ii) upon the completion of the sale of the maximum number of shares under the plan. The aggregate number of shares to be sold under the plan is 992 shares.

Certain of our officers or directors have made elections to participate in, and are participating in, our dividend reinvestment plan and employee stock purchase plan and have made, and may from time to time make, elections to have shares withheld to cover withholding taxes or pay the exercise price of options, which may be designed to satisfy the affirmative defense conditions of Rule 10b5-1 under the Exchange Act or may constitute non-Rule 10b5–1 trading arrangements (as defined in Item 408(c) of Regulation S-K).

ITEM 6. EXHIBITS

Exhibit Number		Description of Document
3.1	—	Sixth Amended and Restated Certificate of Incorporation of Intercontinental Exchange, Inc., effective August 22, 2022 (incorporated by reference to Exhibit 3.1 to Intercontinental Exchange, Inc.'s Current Report on Form 8-K filed with the SEC on August 22, 2022, File No. 001-36198).
3.2	—	Ninth Amended and Restated Bylaws of Intercontinental Exchange, Inc., effective August 22, 2022 (incorporated by reference to Exhibit 3.2 to Intercontinental Exchange, Inc.'s Current Report on Form 8-K filed with the SEC on August 22, 2022, File No. 001-36198).
31.1*	—	Rule 13a-14(a)/15d-14(a) Certification of Chief Executive Officer.
31.2*	—	Rule 13a-14(a)/15d-14(a) Certification of Chief Financial Officer.
32.1**	—	Section 1350 Certification of Chief Executive Officer.
32.2**	—	Section 1350 Certification of Chief Financial Officer.
101	—	The following materials from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended June 30, 2023, formatted in Inline XBRL (Extensible Business Reporting Language): (i) the Consolidated Balance Sheets, (ii) the Consolidated Statements of Income, (iii) the Consolidated Statements of Comprehensive Income, (iv) the Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest (v) Consolidated Statements of Cash Flows and (vi) Notes to Consolidated Financial Statements, tagged as blocks of text.
104	—	The cover page from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended June 30, 2023 formatted in Inline XBRL.

* Filed herewith.

** Furnished herewith. These exhibits shall not be deemed “filed” for purposes of Section 18 of the Securities Exchange Act of 1934, or otherwise subject to the liability of that Section. Such exhibits shall not be deemed incorporated into any filing under the Securities Act of 1933 or the Securities Exchange Act of 1934.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this Report to be signed on its behalf by the undersigned, thereunto duly authorized.

Intercontinental Exchange, Inc.
(Registrant)

Date: August 3, 2023

By: /s/ A. Warren Gardiner
A. Warren Gardiner
Chief Financial Officer
(Principal Financial Officer)